

Sequential Modeling Architecture - II

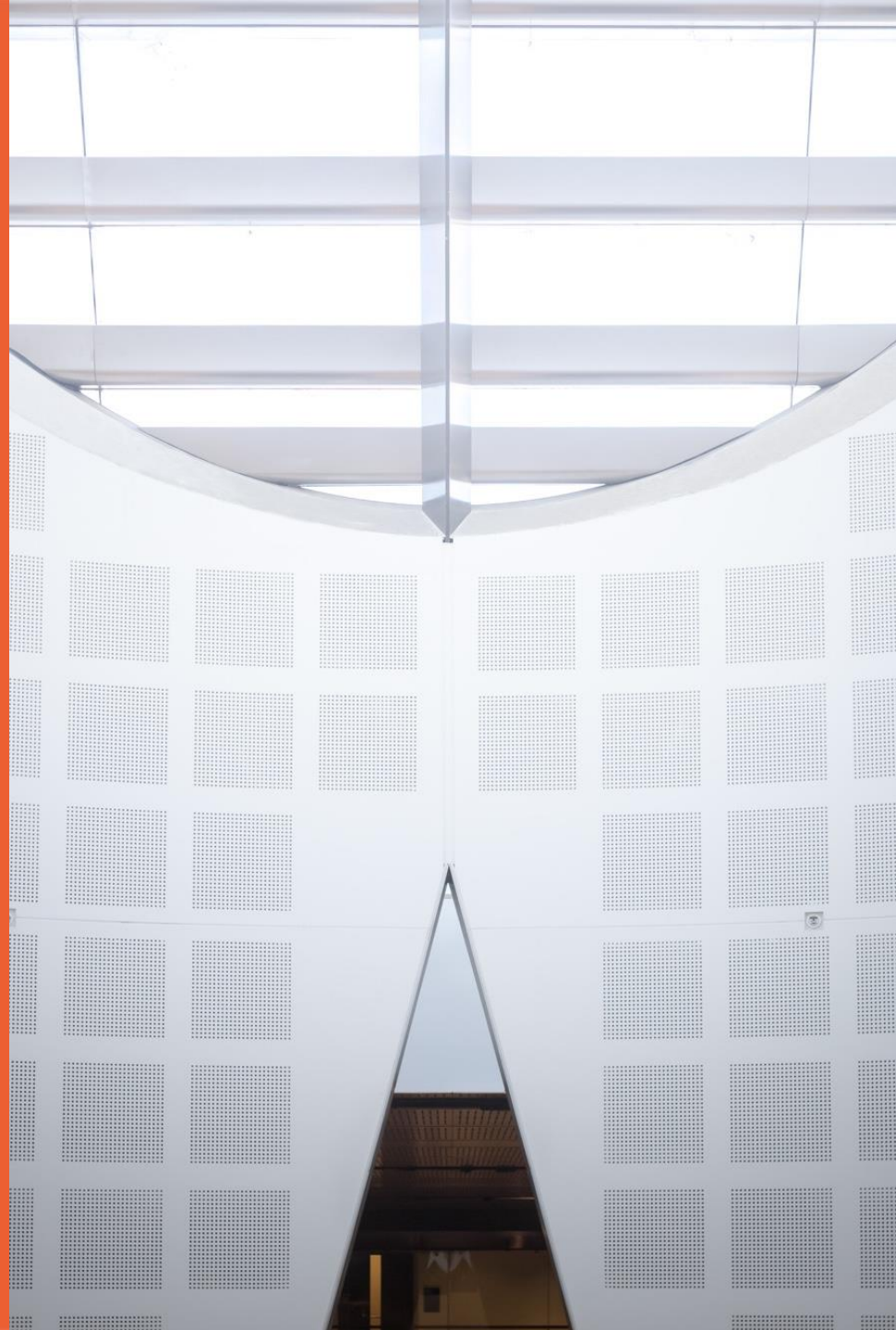
--BERT, GPT, Mamba

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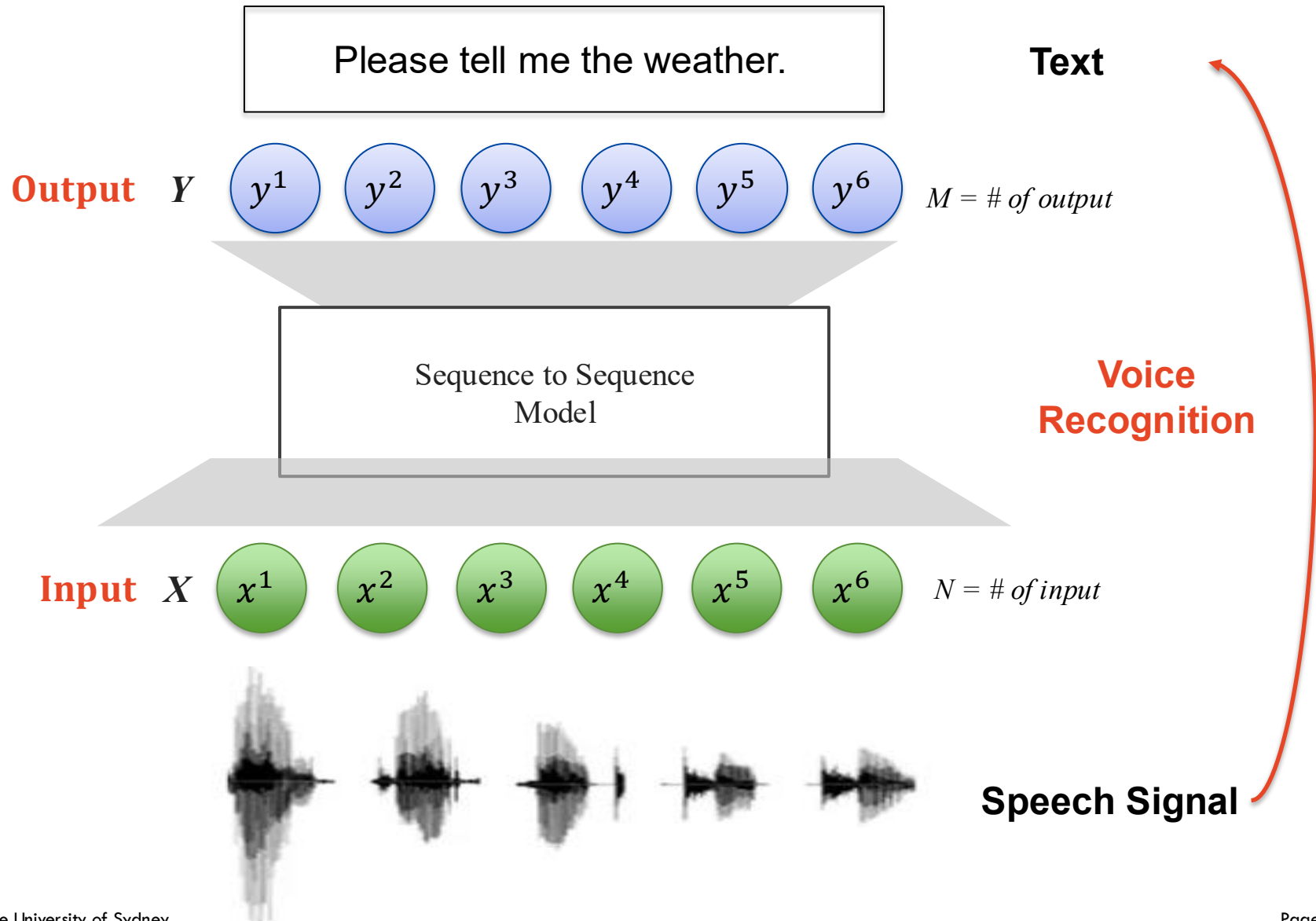
THE UNIVERSITY OF
SYDNEY



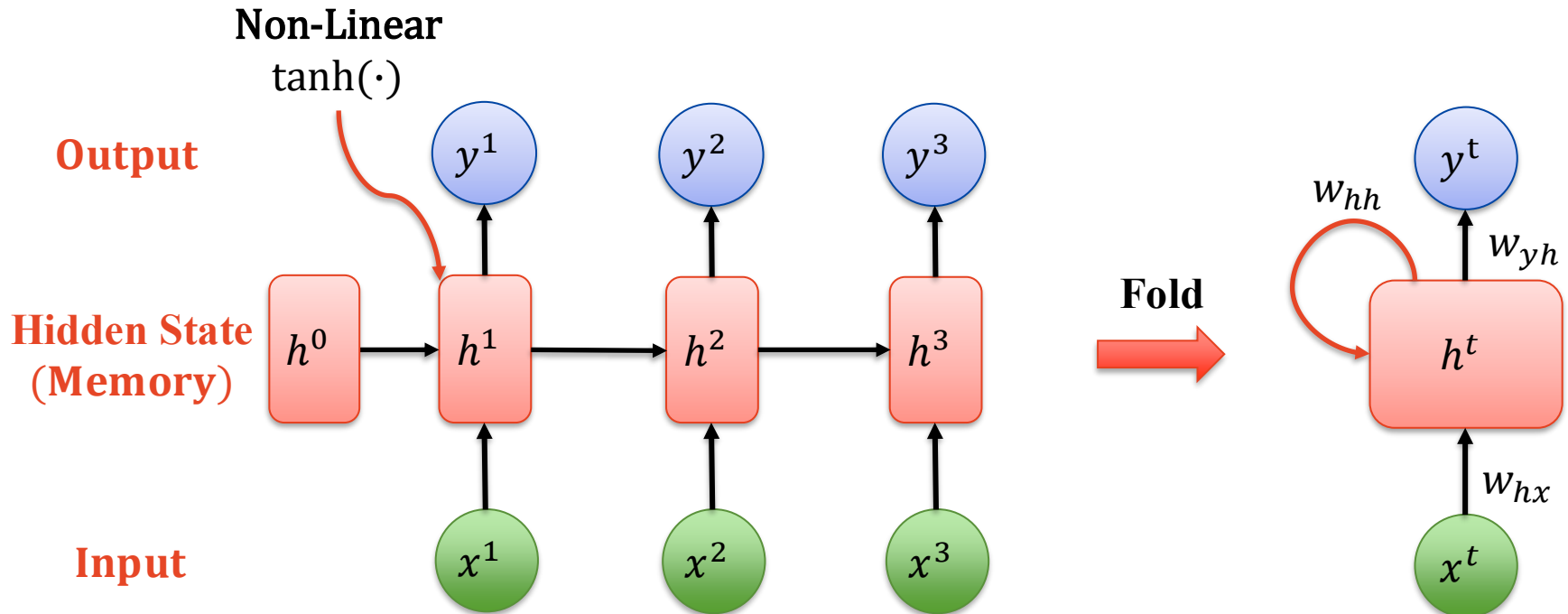
Quick Review

S2S Model & RNN & Transformer

Sequence to Sequence Model



Recurrent Neural Network



$$\square h_t = \tanh(w_{hh}h_{t-1} + w_{hx}x_t + b_h)$$

$$\square y_t = w_{yh}h_t$$

- Memory Efficient
- Infinite Context Window
- Not parallelizable

Recurrent Neural Network

❑ Vanishing (exploding) gradient problem

❑ $h_t = \tanh(w_{hh}h_{t-1} + w_{hx}x_t + b_h)$

❑ $y_t = w_{yh}h_t$

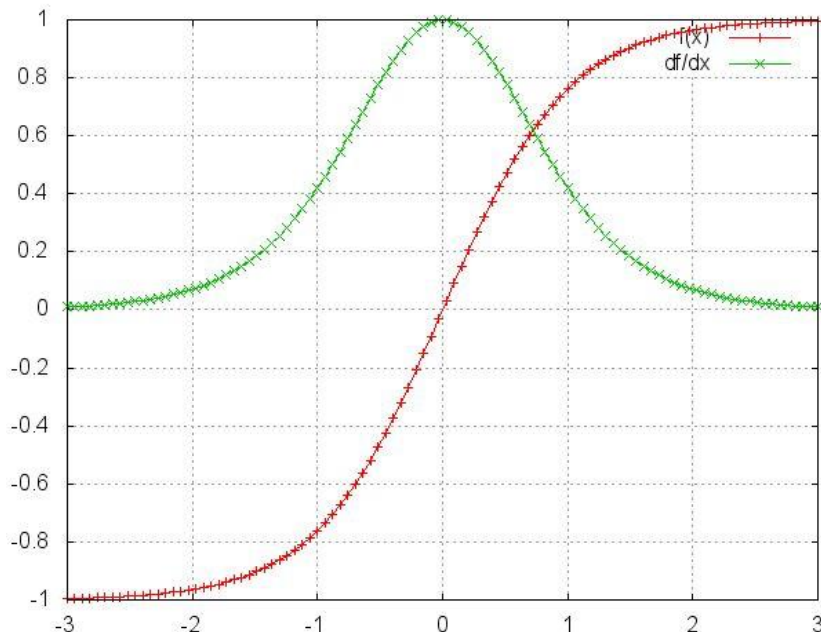
❑ For every step, its loss is E_n

$$0.9^{1000} \approx 0$$

$$1.01^{1000} \approx 21000$$

$$\frac{\partial E_3}{\partial w_{hh}} = \frac{\partial E_3}{\partial y_3} \frac{\partial y_3}{\partial h_3} \frac{\partial h_3}{\partial w_{hh}}$$

$$(\tanh)'w_{hh}(\tanh)'w_{hh}(\tanh)'w_{hh} \cdots$$



$$(\tanh)' \leq 1$$

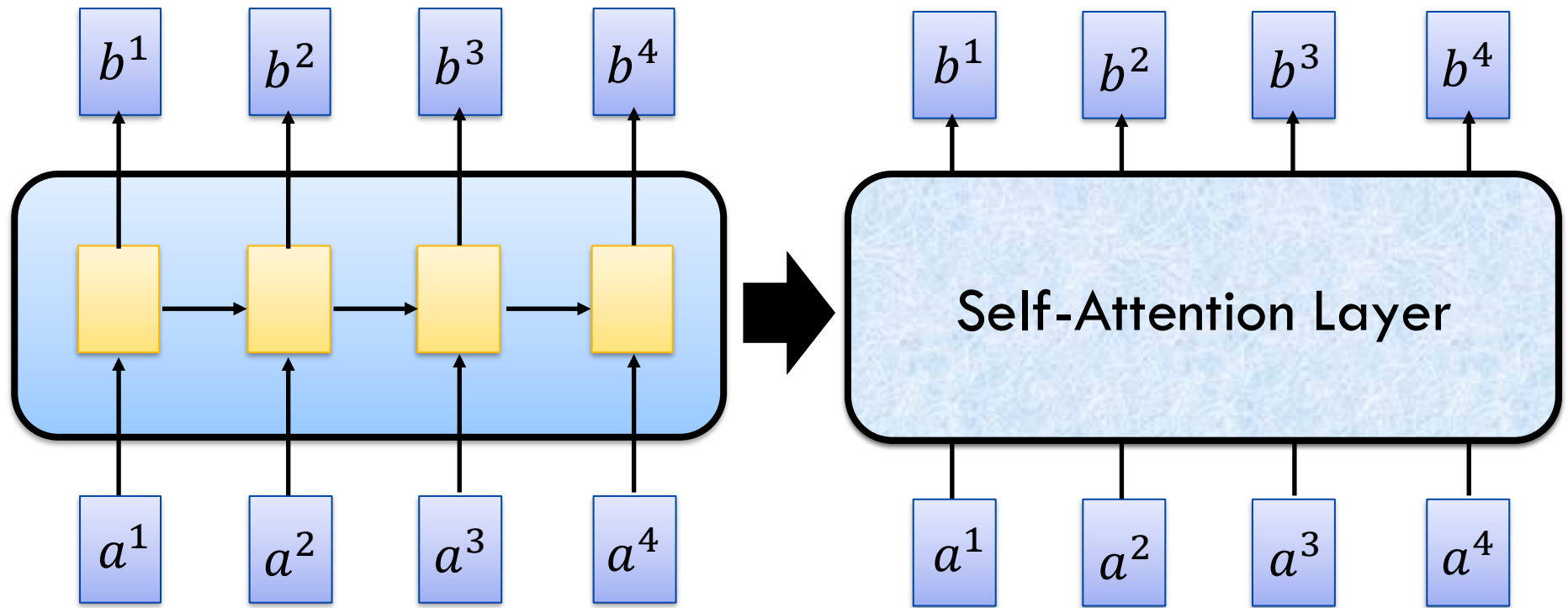
If $0 < w_{hh} < 1$, **Vanishing Gradients**

$$(\tanh)'w_{hh}(\tanh)'w_{hh}(\tanh)'w_{hh} \cdots \rightarrow 0$$

If w_{hh} is larger, **Exploding Gradients**

$$(\tanh)'w_{hh}(\tanh)'w_{hh}(\tanh)'w_{hh} \cdots \rightarrow \infty$$

Self-Attention



You can try to replace any thing that has been done by RNN with self-attention.

Attention Is All You Need

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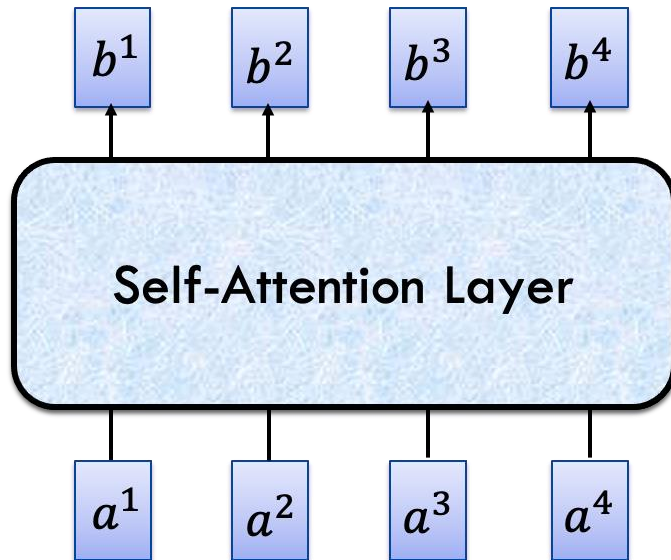
Lukasz Kaiser*
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Attention is all
you need.

Self-Attention



q : **query** (to match others)

$$q^i = W^q a^i$$

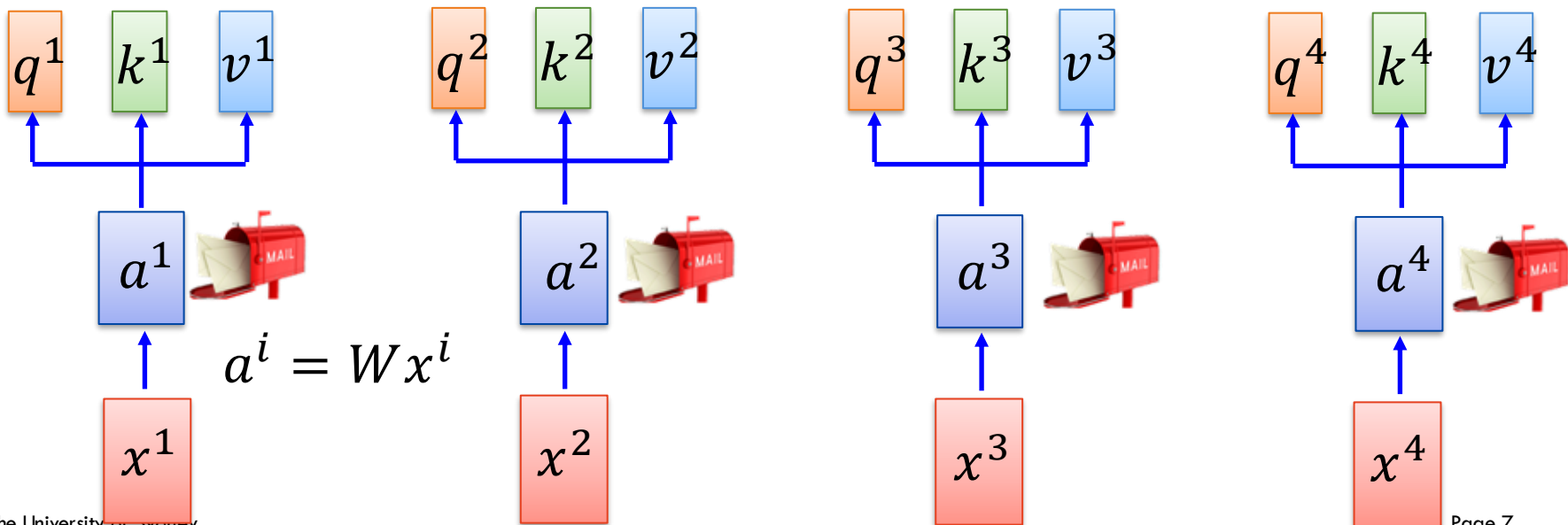


k : **key** (to be matched)

$$k^i = W^k a^i$$

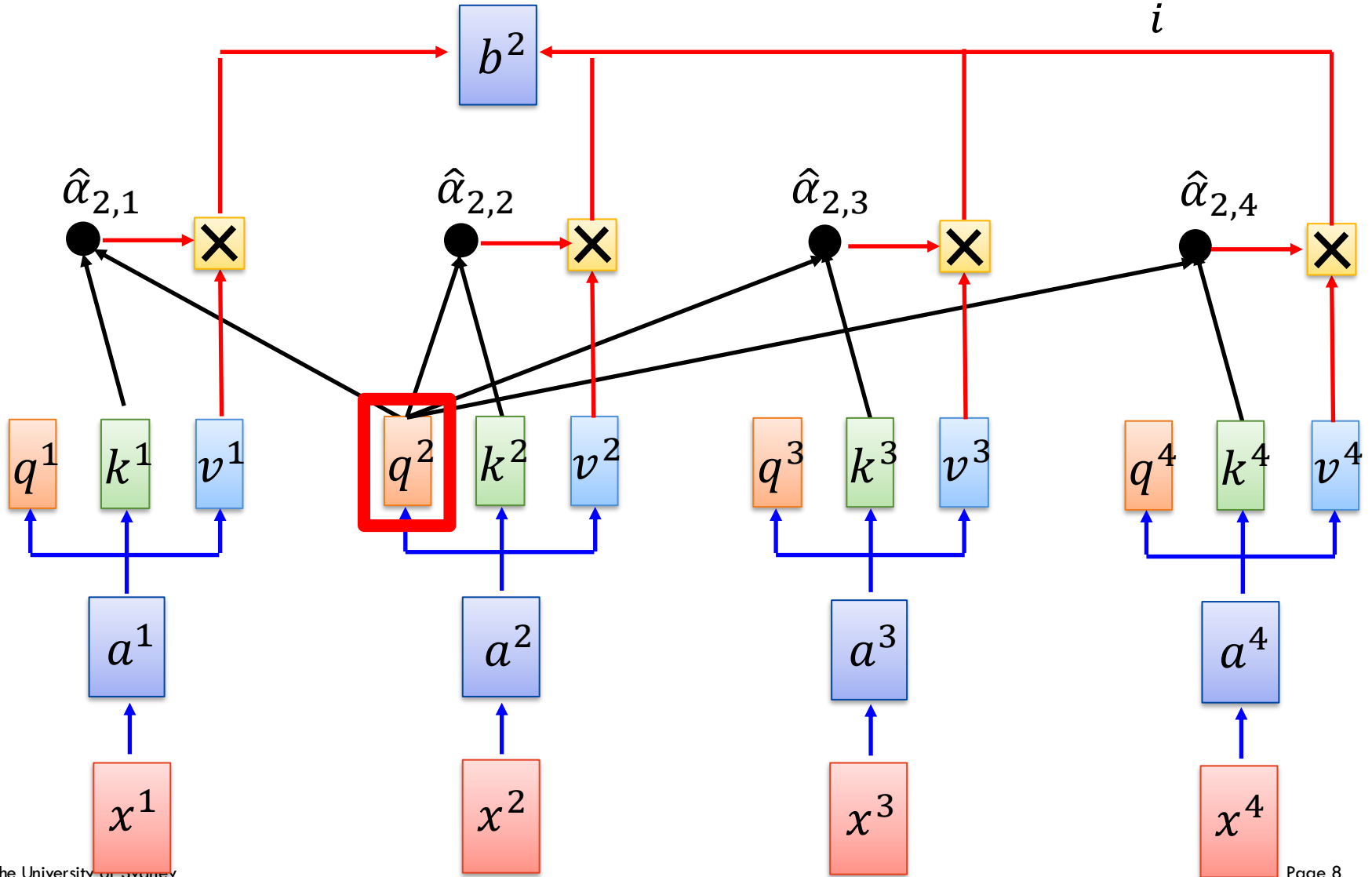
v : **value** (information to be extracted)

$$v^i = W^v a^i$$



Self-Attention

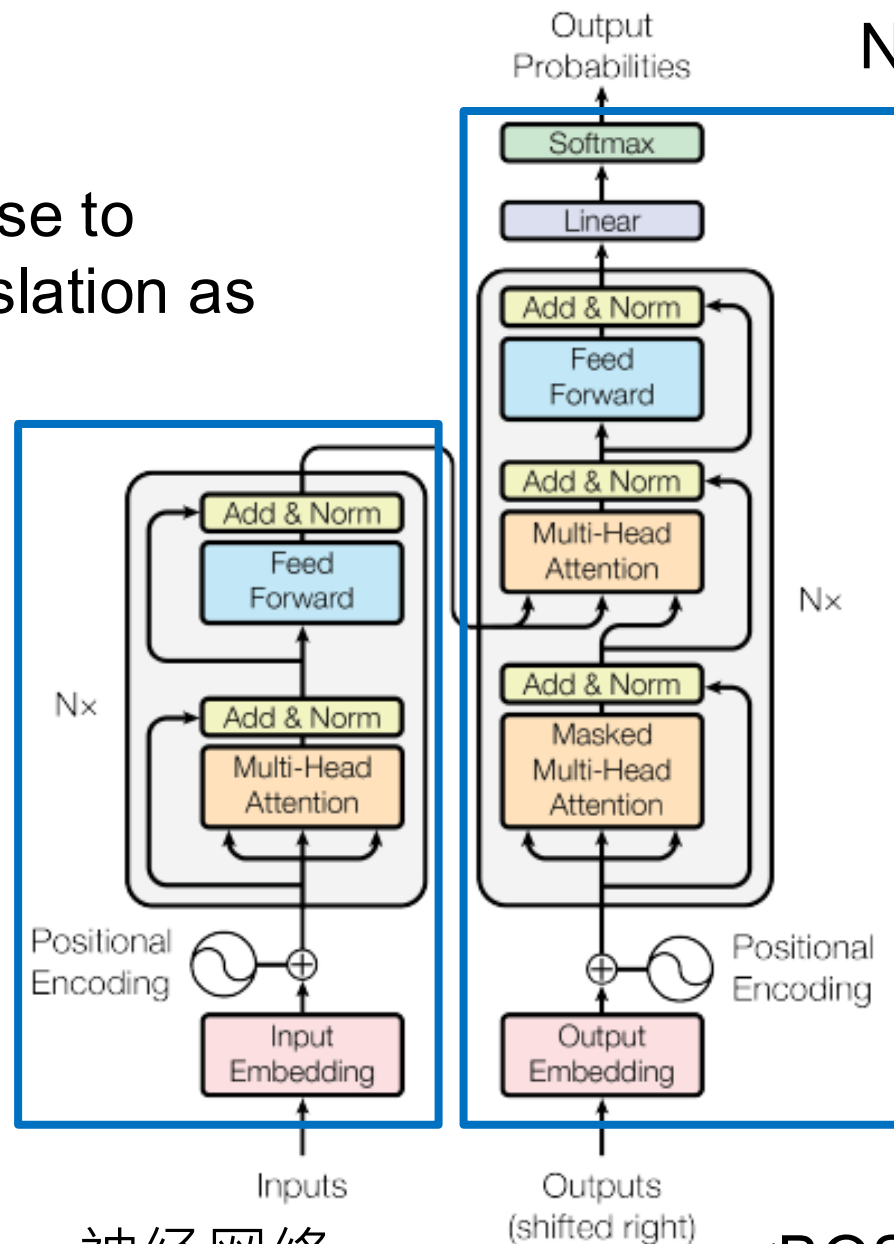
$$b^2 = \sum_i \hat{\alpha}_{2,i} v^i$$



Transformer

Using Chinese to
English translation as
example

Encoder



Neural Network

Decoder

神经网络

<BOS>

BERT

A **transformer** uses Encoder stack to model input, and uses Decoder stack to model output (using input information from encoder side).

If we are only interested in training a language model for the input for some other tasks, then we do not need the Decoder of the transformer, that gives us **BERT**.

1-of-N Encoding

apple = [1 0 0 0 0]

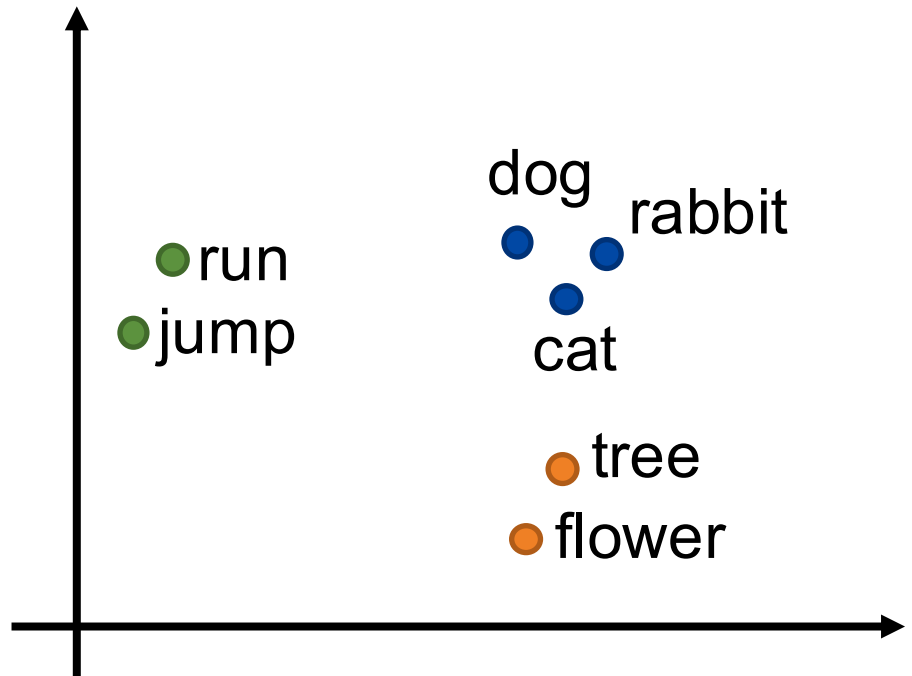
bag = [0 1 0 0 0]

cat = [0 0 1 0 0]

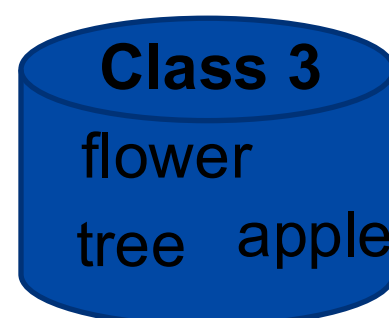
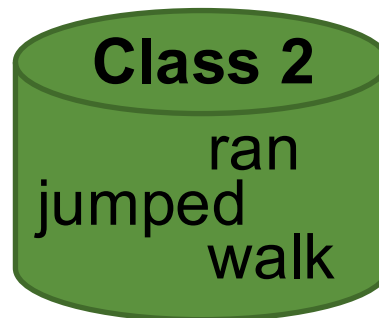
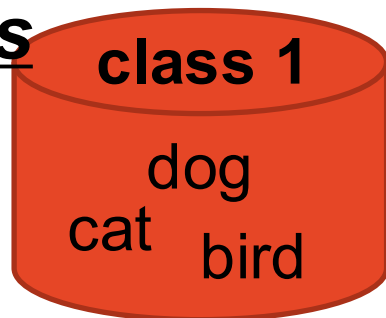
dog = [0 0 0 1 0]

elephant = [0 0 0 0 1]

Word Embedding



Word Class



A word can have multiple meanings.

Have you paid that money to the **bank** yet ?

It is safest to deposit your money in the **bank** .

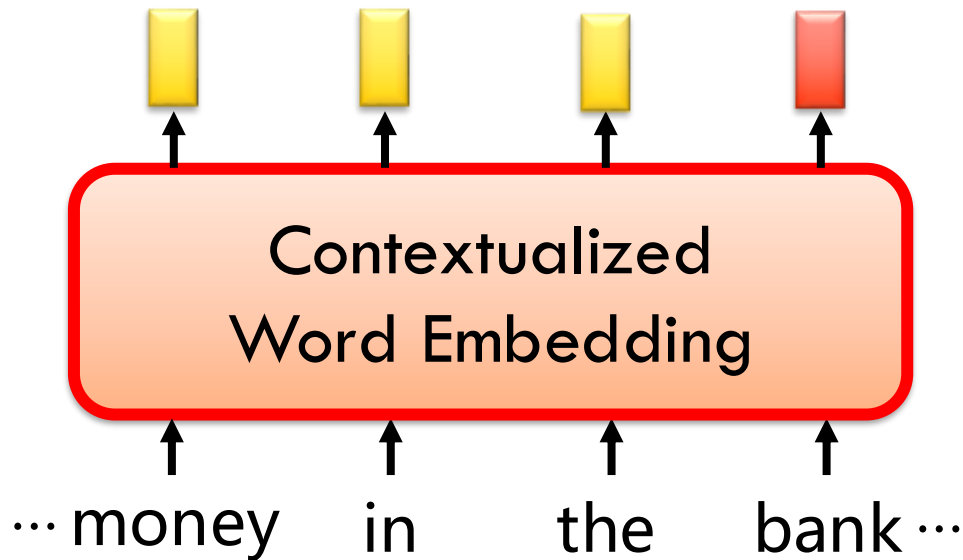
The victim was found lying dead on the river **bank** .

They stood on the river bank to fish.

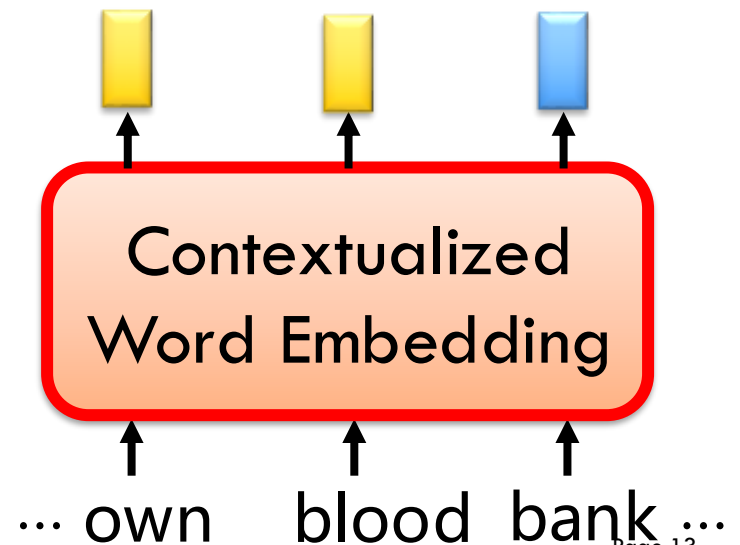
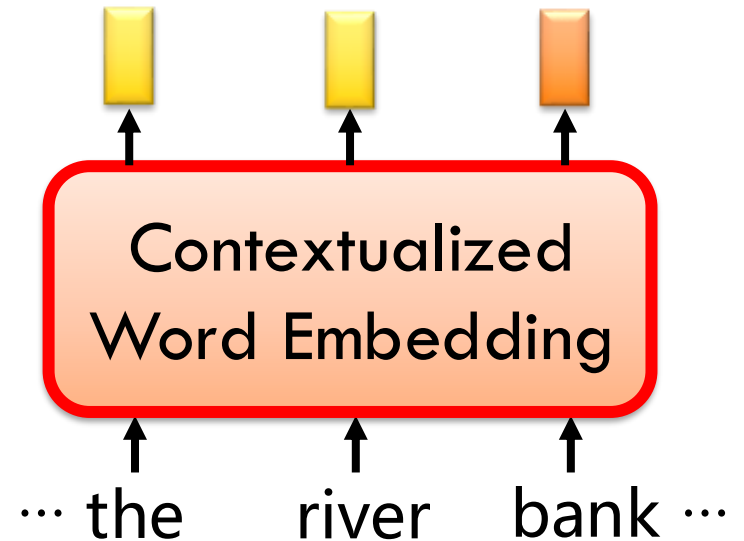
The hospital has its own blood **bank**.

The third sense or not?

Contextualized Word Embedding



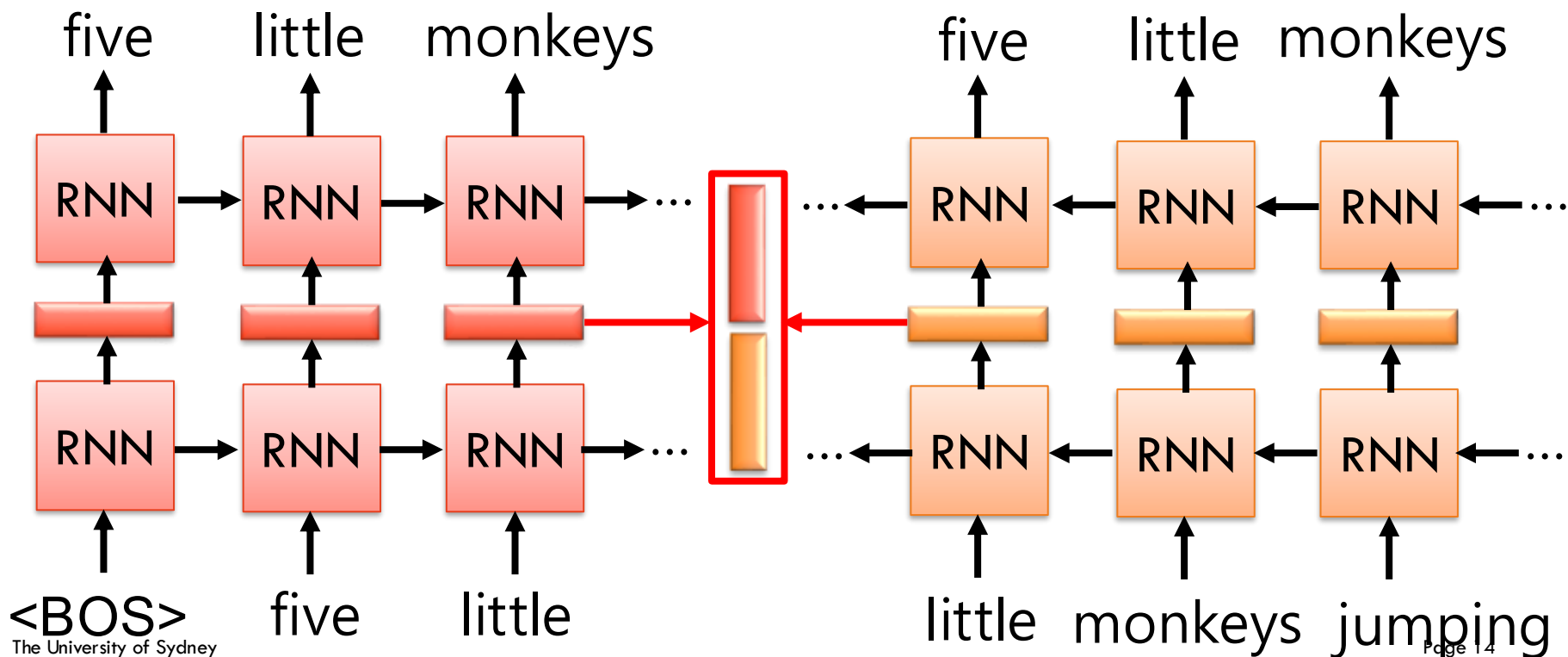
Instead of assigning a fixed vector to each word, we generate a different representation depending on the context.



Embeddings from Language Model (ELMO)



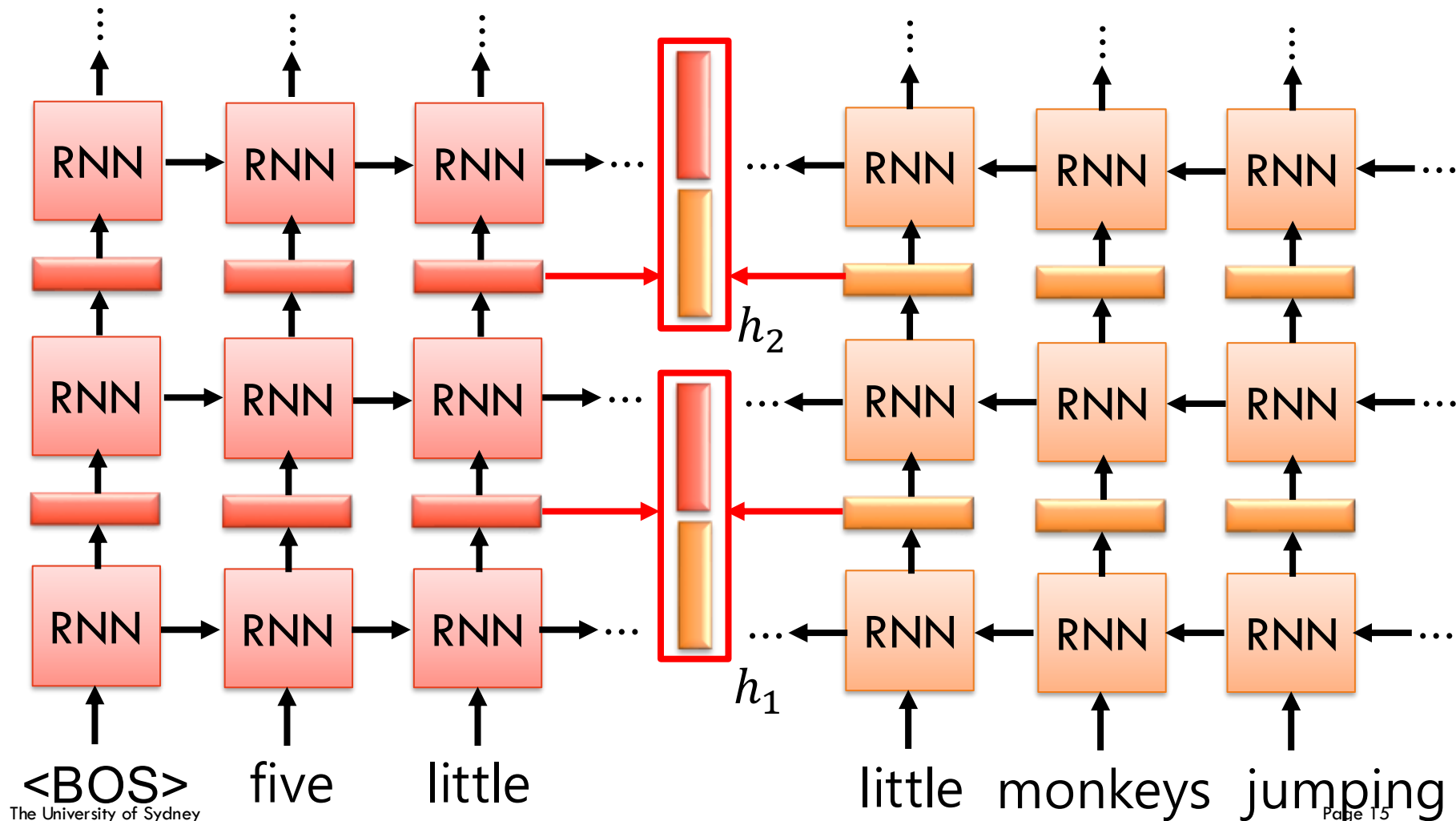
- RNN-based language models (trained from lots of sentences)
e.g. given “five little monkeys jumping on the bed”



ELMO

Each layer in deep RNN can generate a latent representation.

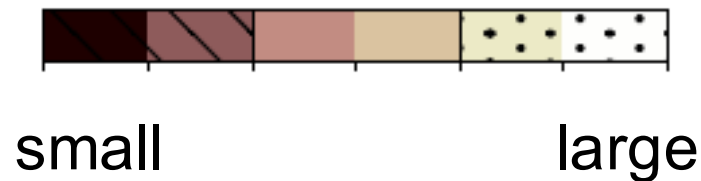
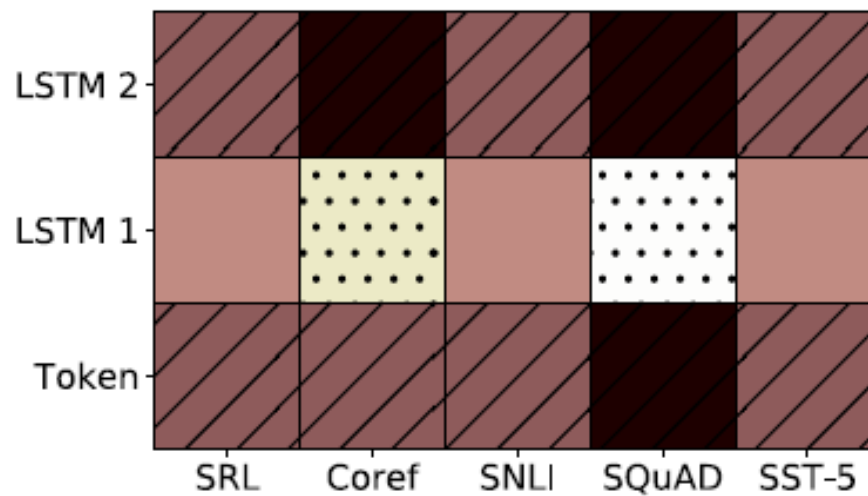
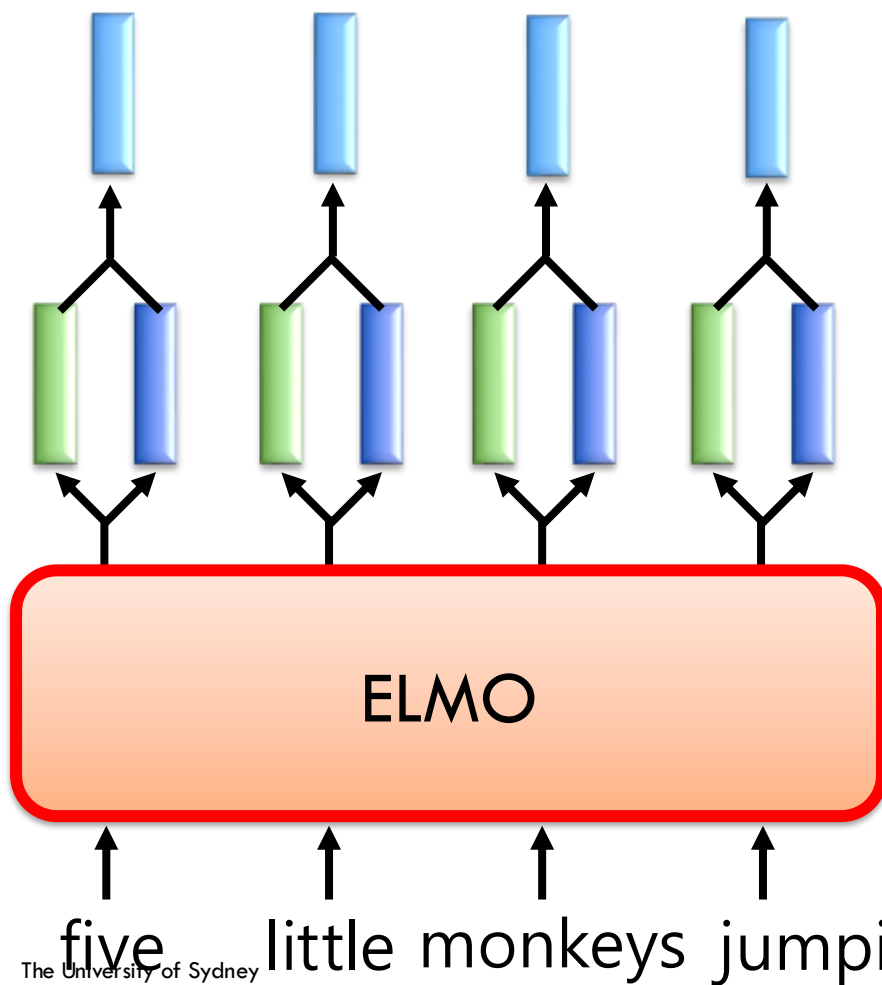
Which one should we use???



ELMO

$$\text{Blue Bar} = \alpha_1 \text{ Green Bar} + \alpha_2 \text{ Blue Bar}$$

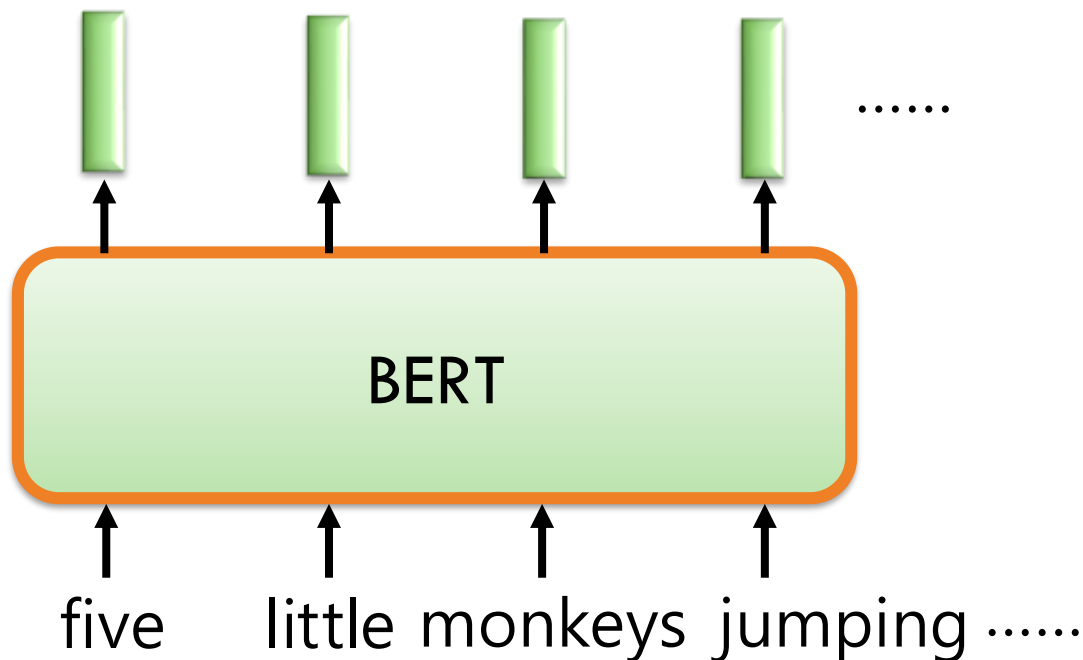
Learned with the down stream tasks



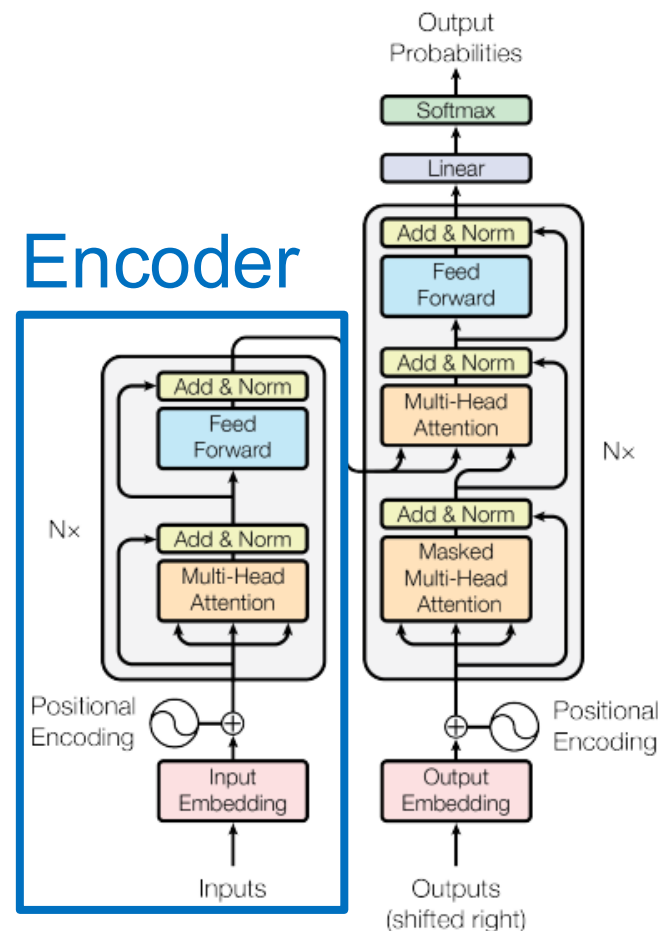
Bidirectional Encoder Representations from Transformers (BERT)

- BERT = Encoder of Transformer

Learned from a large amount of text without annotation

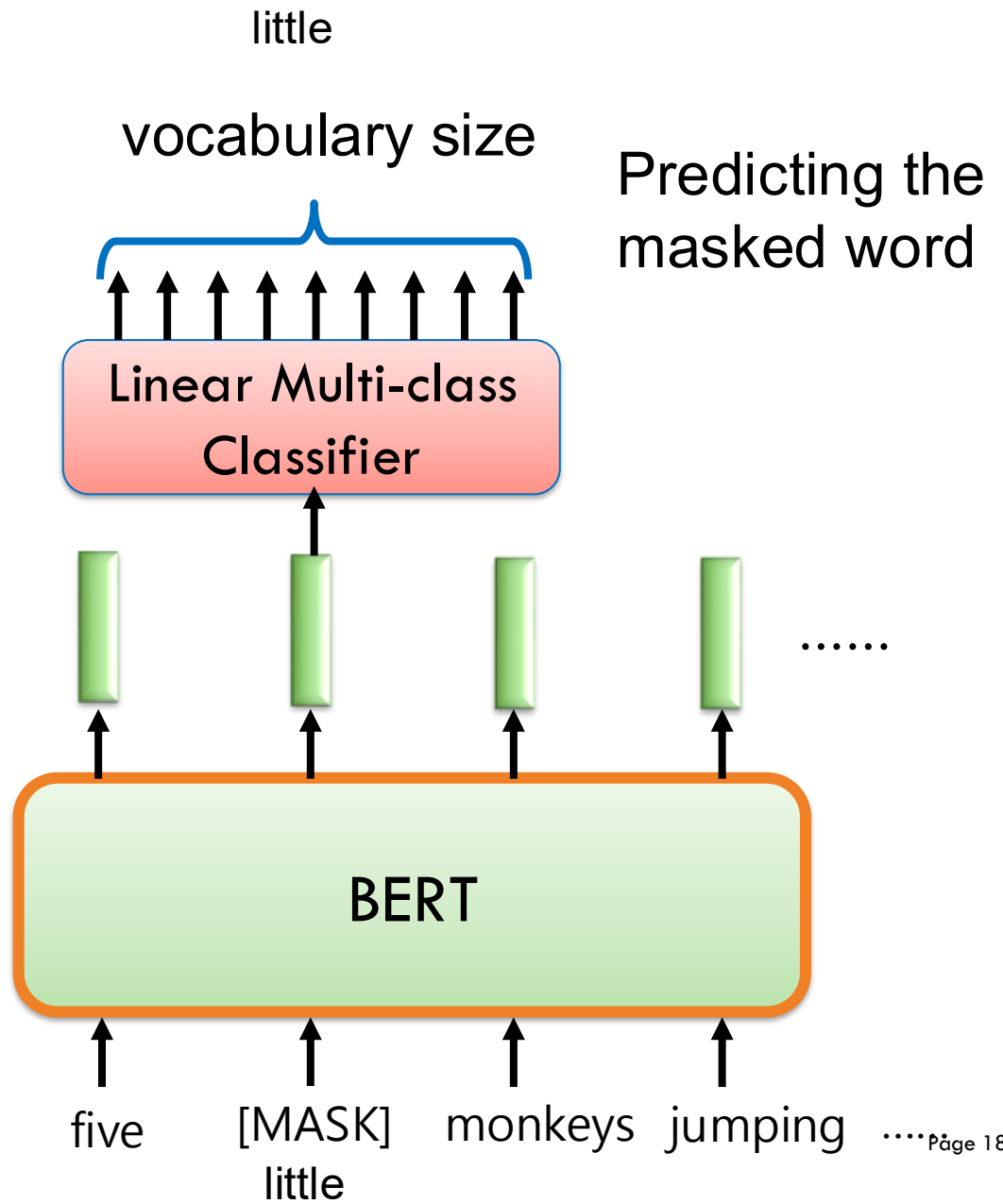


Encoder



Training of BERT

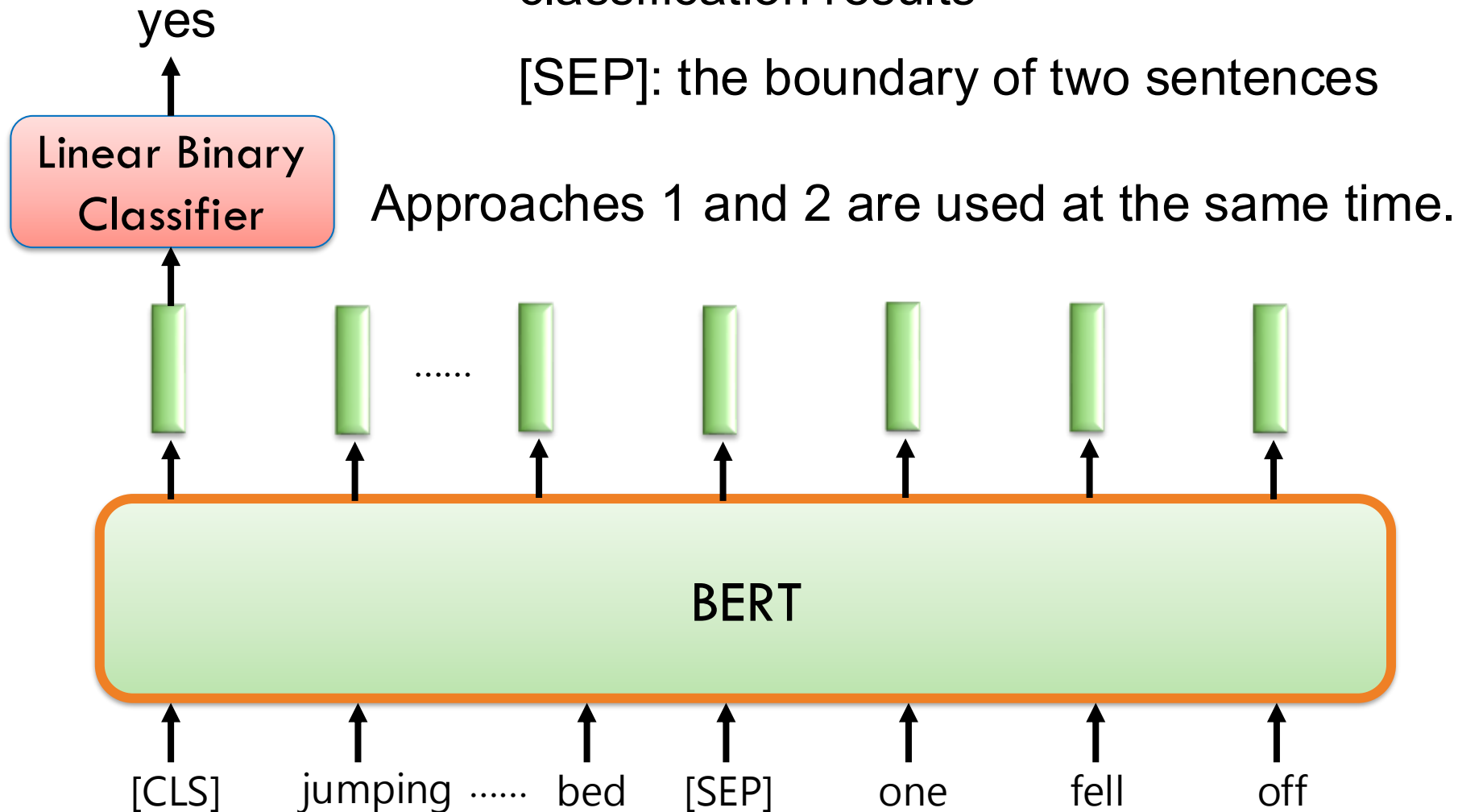
- Approach 1:
Masked LM



Training of BERT Approach 2: Next Sentence Prediction

[CLS]: the position that outputs classification results

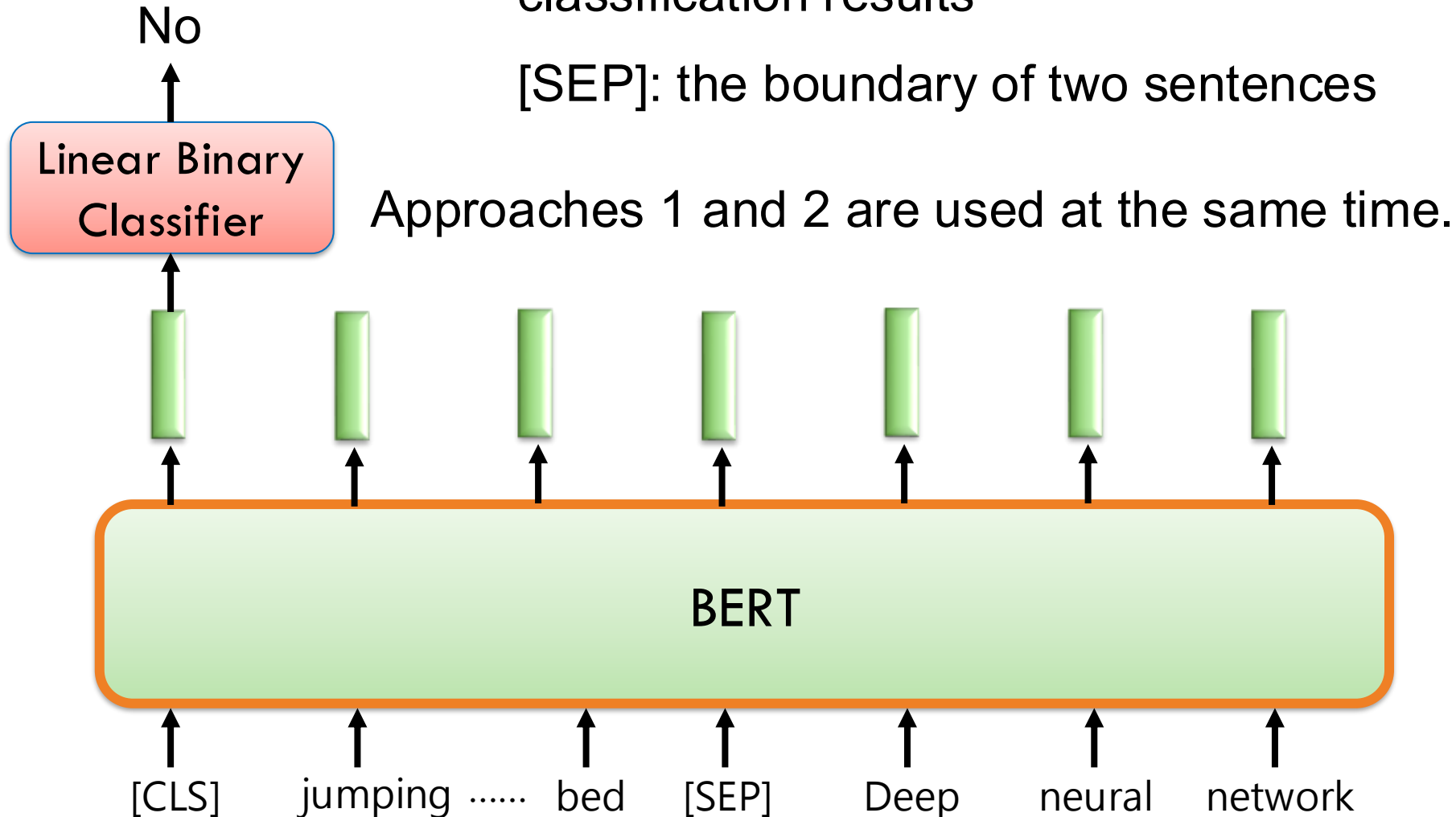
[SEP]: the boundary of two sentences



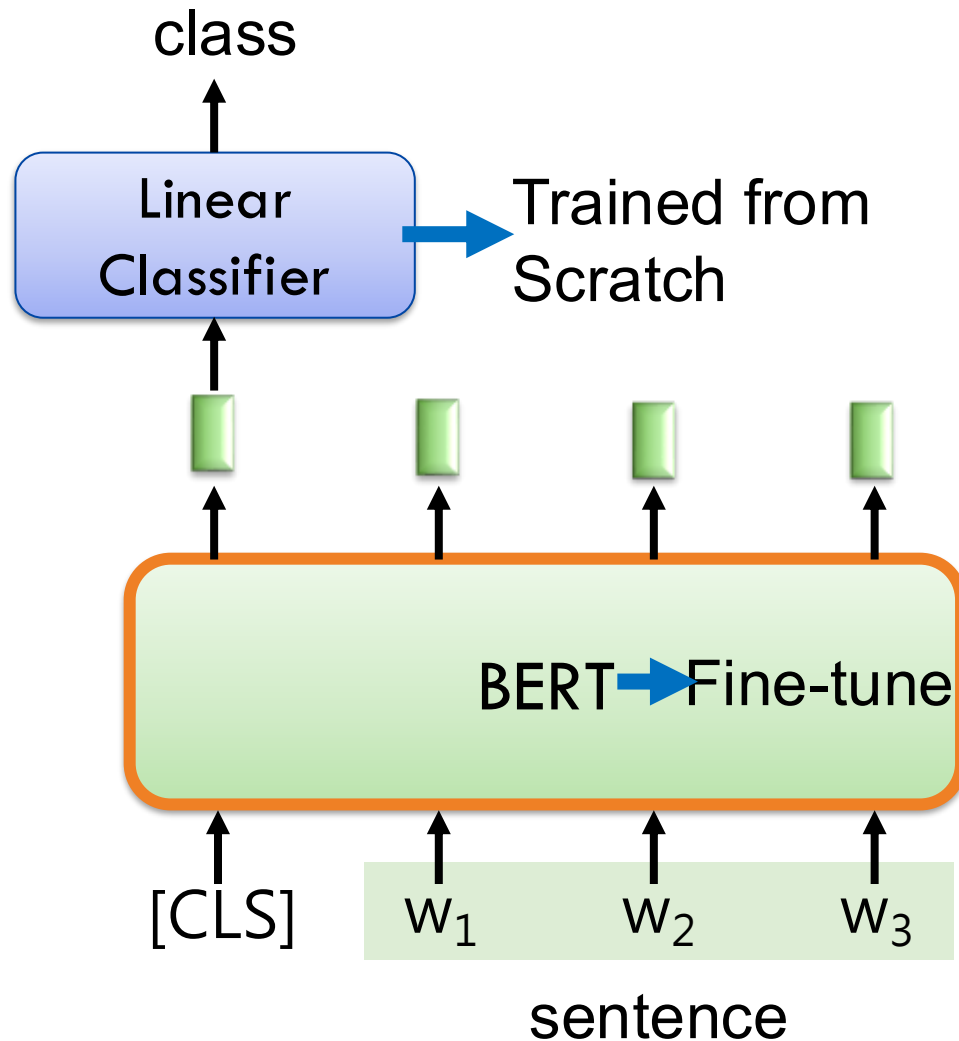
Training of BERT Approach 2: Next Sentence Prediction

[CLS]: the position that outputs classification results

[SEP]: the boundary of two sentences



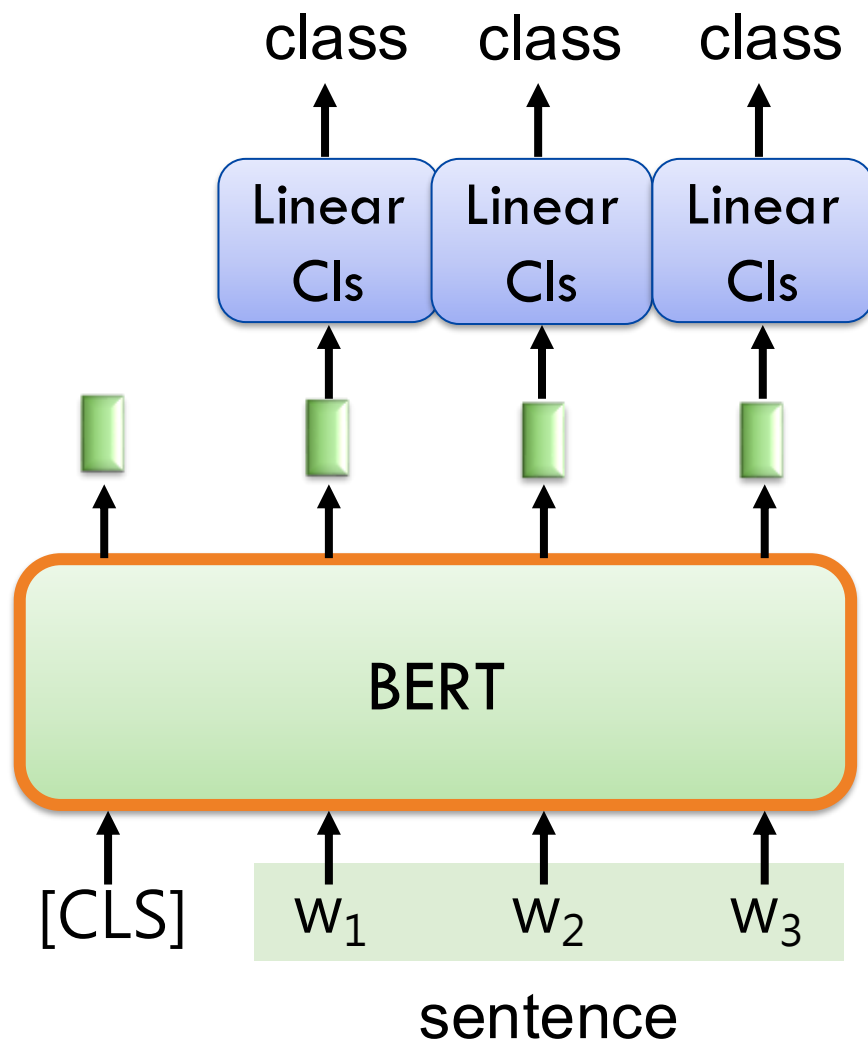
How to use BERT – Case 1



Input: single sentence,
output: class

Example:
Sentiment analysis
Document Classification

How to use BERT – Case 2



Input: single sentence,
output: class of each word

Example: Slot filling

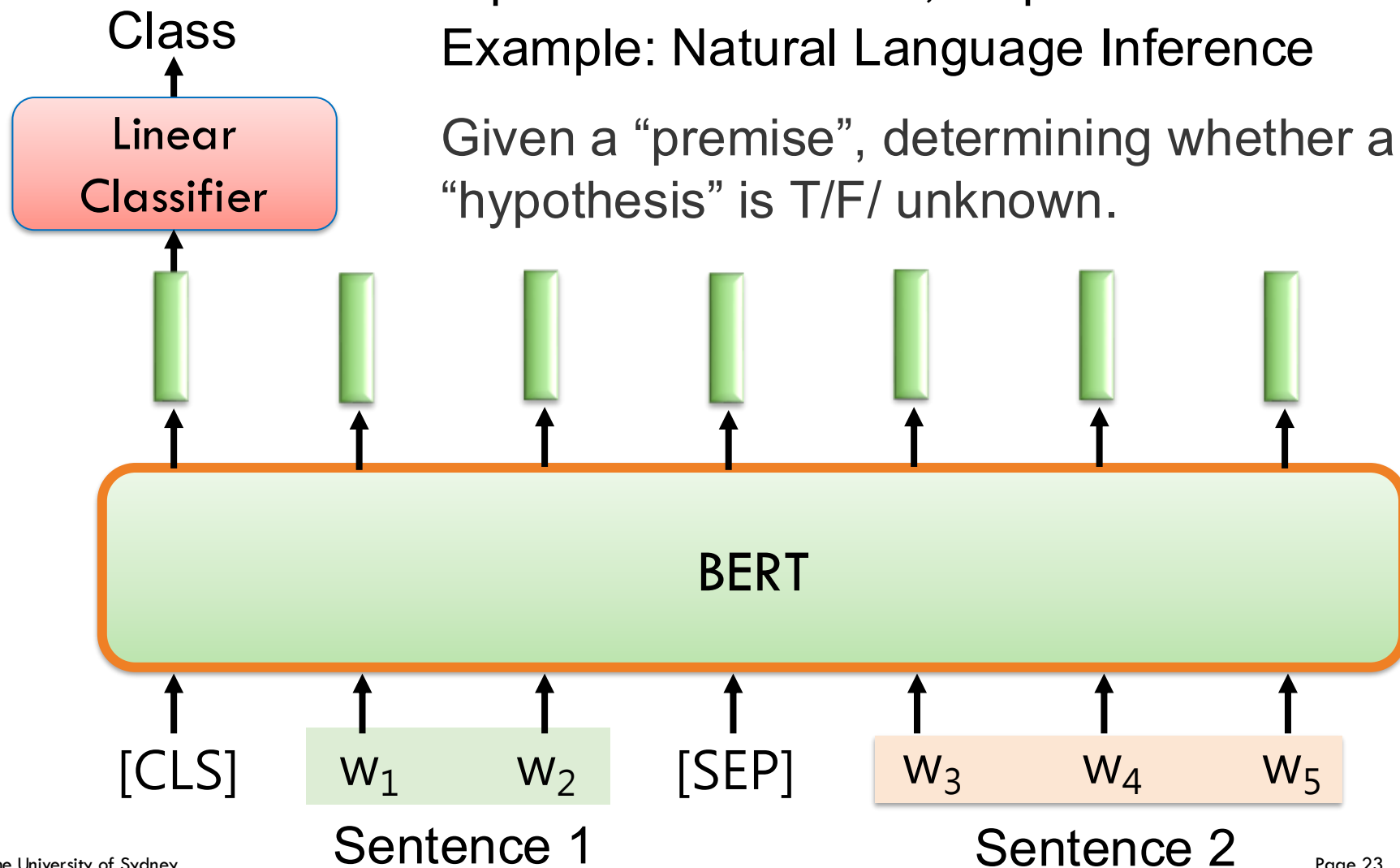


How to use BERT – Case 3

Input: two sentences, output: class

Example: Natural Language Inference

Given a “premise”, determining whether a “hypothesis” is T/F/ unknown.

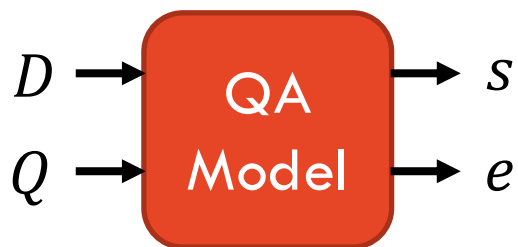


How to use BERT – Case 4

- Extraction-based Question Answering (QA) (E.g. SQuAD)

Document: $D = \{d_1, d_2, \dots, d_N\}$

Query: $Q = \{q_1, q_2, \dots, q_N\}$



output: two integers (s, e)

Answer: $A = \{d_s, \dots, d_e\}$

In meteorology, precipitation is any product of the condensation of 17 spheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain 77 at 79 locations are called “showers”.

What causes precipitation to fall?

gravity

$s = 17, e = 17$

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?

graupel

Where do water droplets collide with ice crystals to form precipitation?

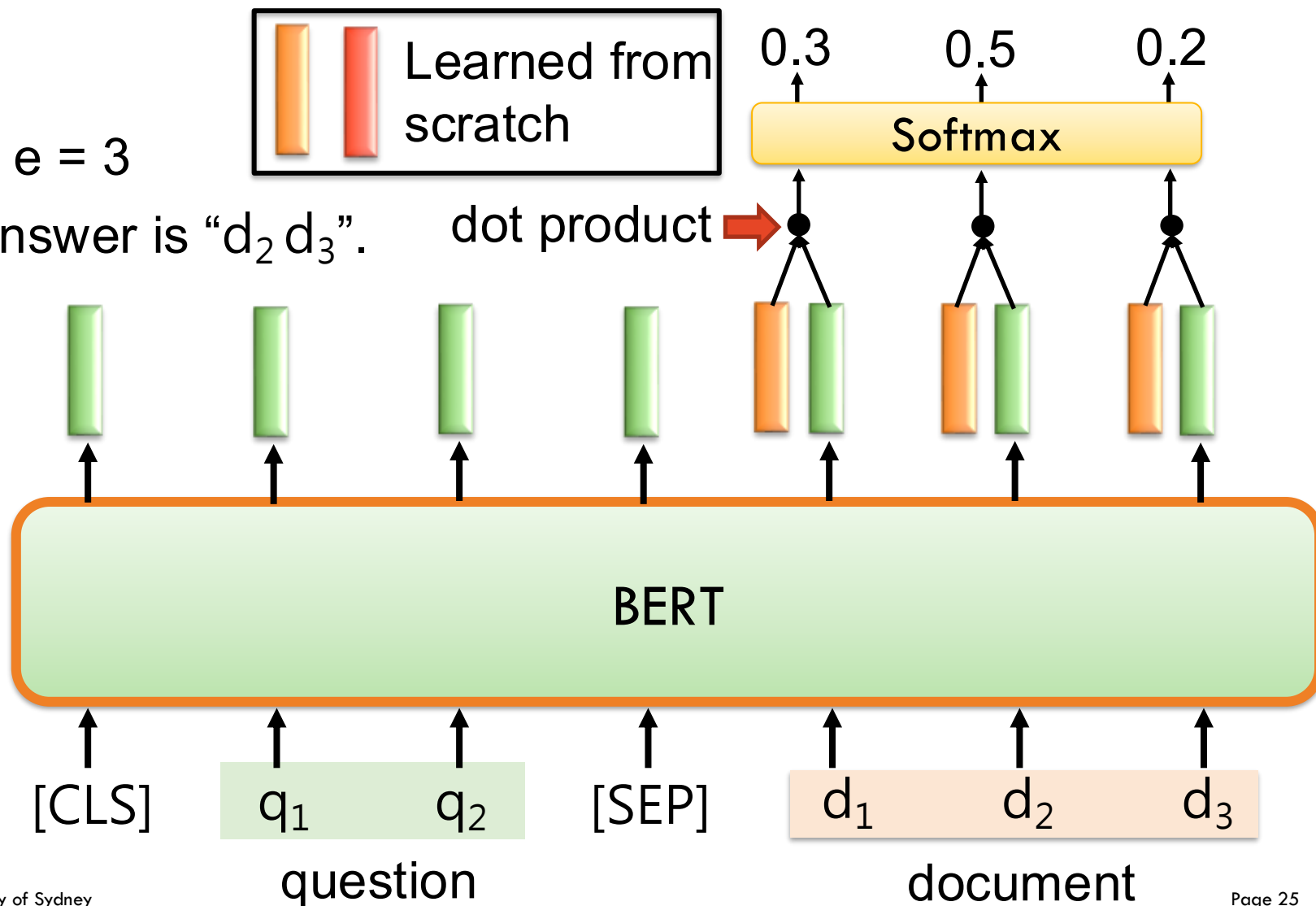
within a cloud

$s = 77, e = 79$

How to use BERT – Case 4

$s = 2, e = 3$

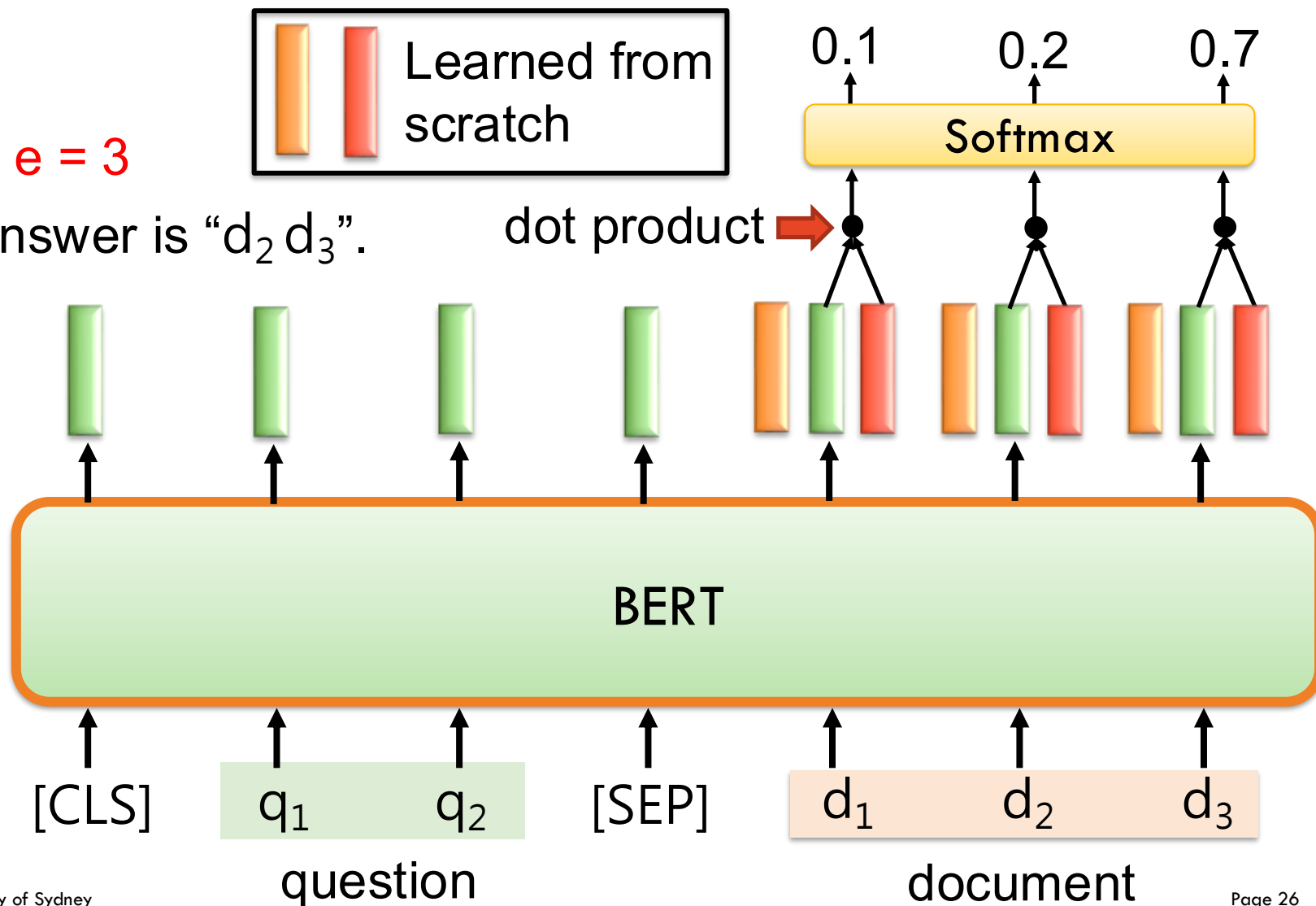
The answer is “ $d_2 d_3$ ”.



How to use BERT – Case 4

$s = 2$, $e = 3$

The answer is “ $d_2 d_3$ ”.

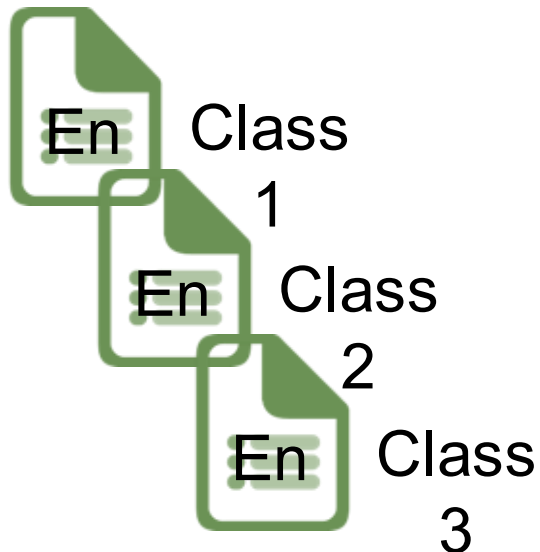


Multilingual BERT

Trained on 104 languages



Task specific training
data for English



Task specific testing
data for Chinese



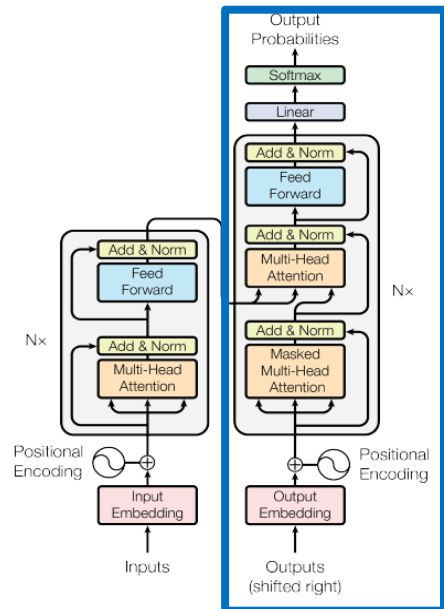
GPT

A **transformer** uses Encoder stack to model input, and uses Decoder stack to model output (using input information from encoder side).

If we are only interested in training a language model for the input for some other tasks, then we do not need the Decoder of the transformer, that gives us **BERT**.

But if we do not have input, we just want to model the “next word”, we can get rid of the Encoder side of a transformer and output “next word” one by one. This gives us **GPT**.

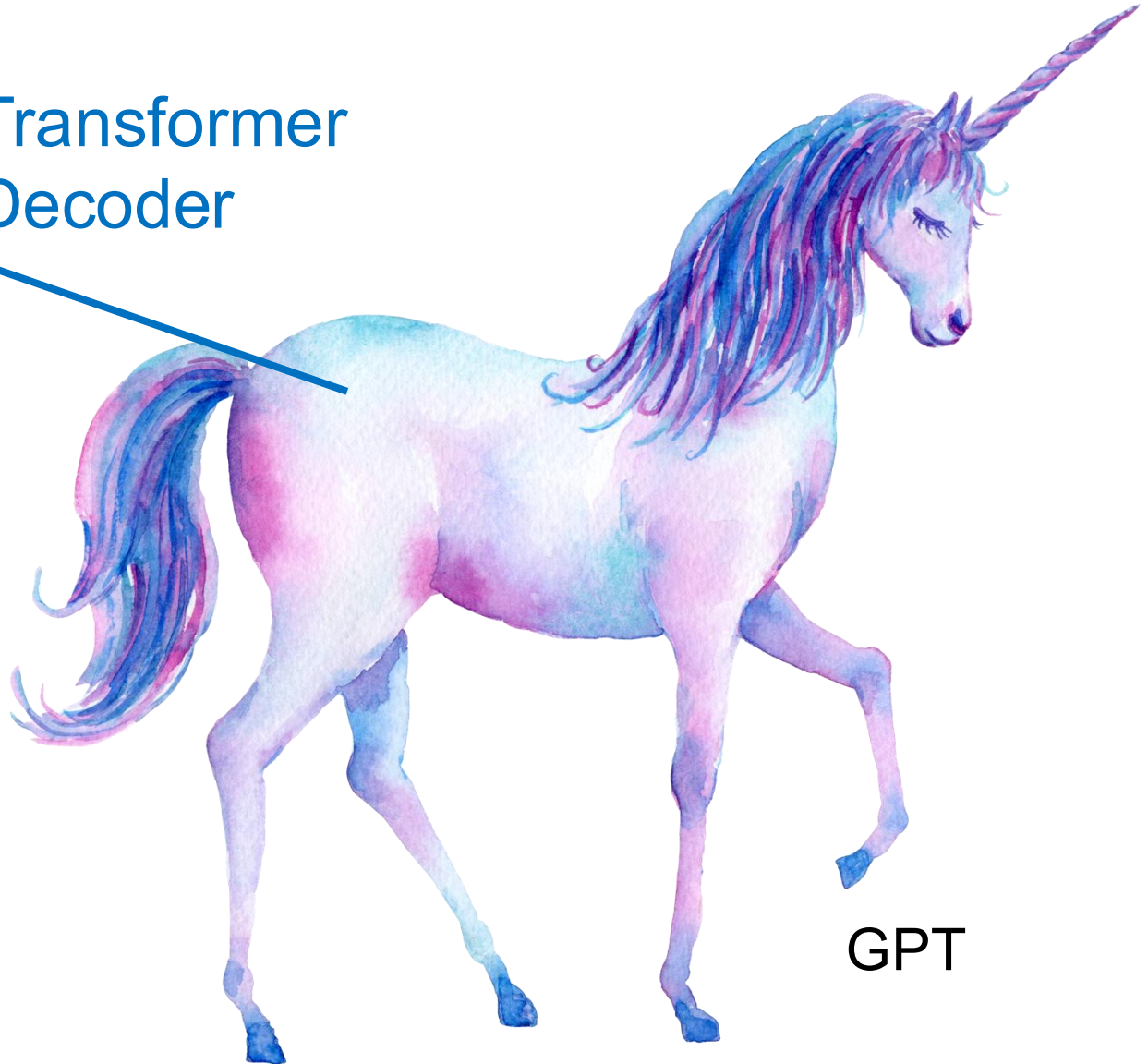
Generative Pre-Training (GPT)



Transformer
Decoder

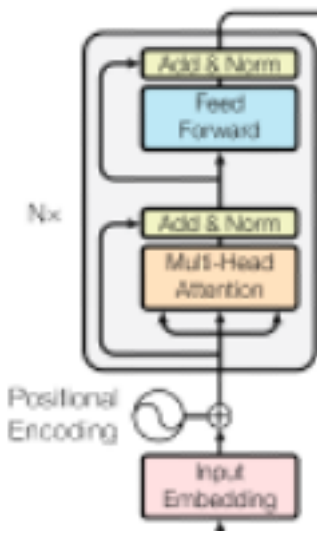
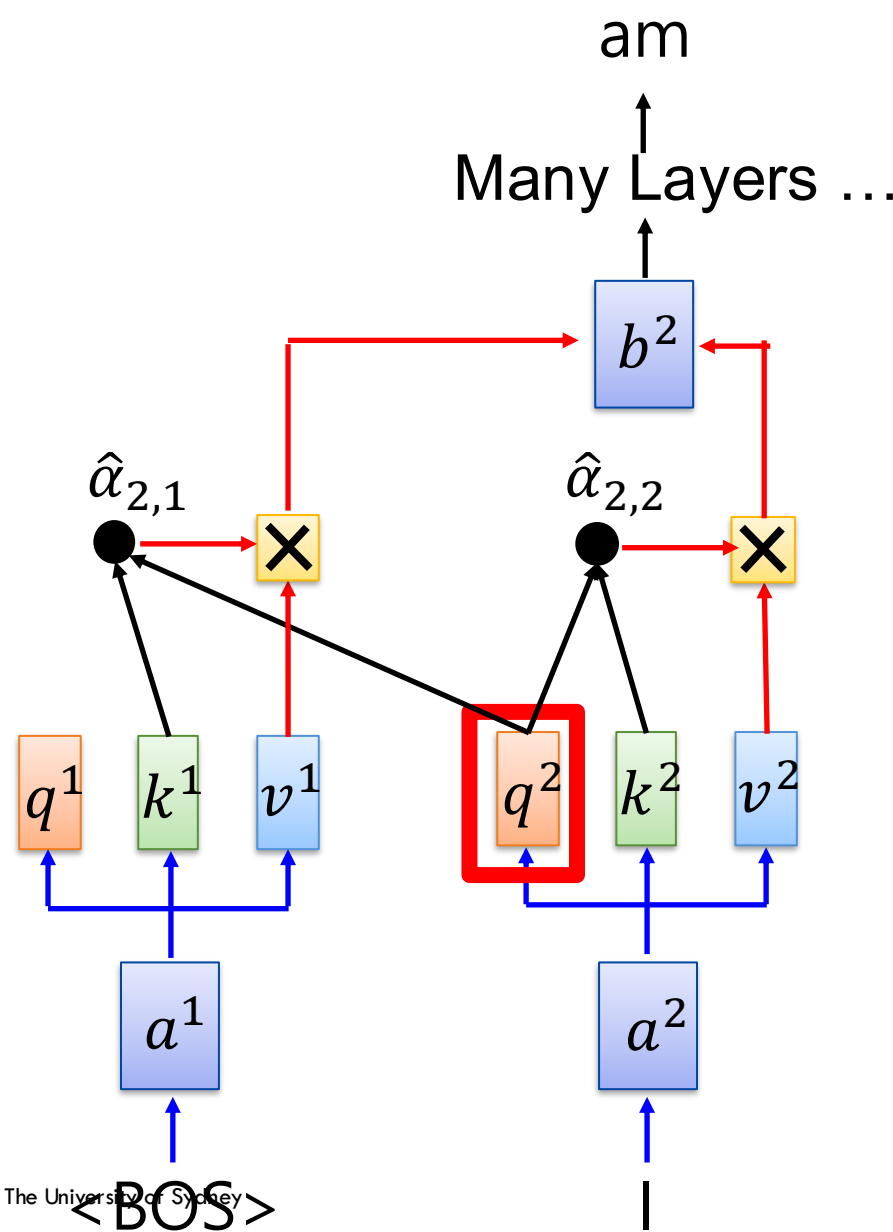
BERT
(340M)

ELMO
(94M)



GPT

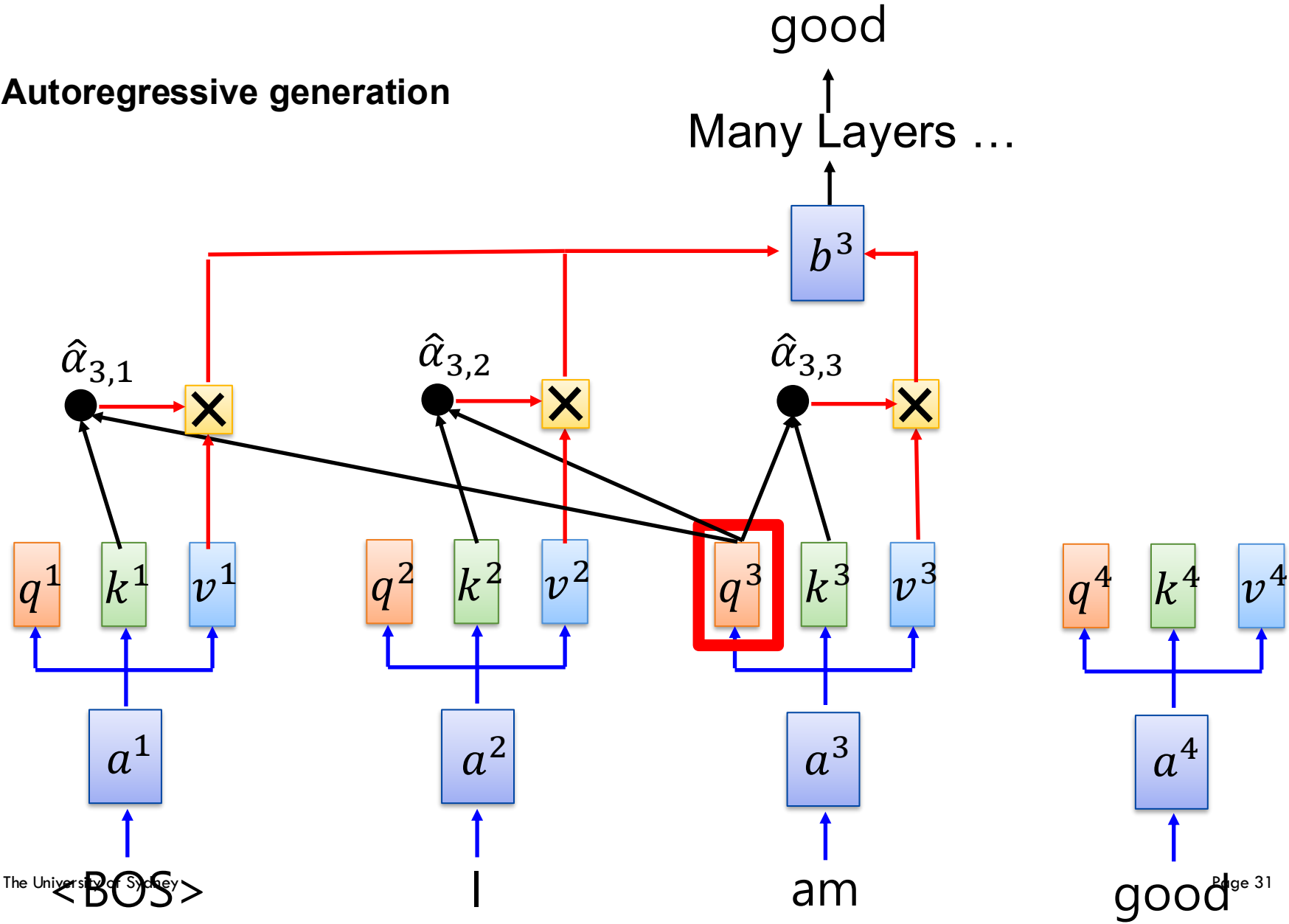
Generative Pre-Training (GPT)



“How are you ?” send into
Transformer Encoder

Generative Pre-Training (GPT)

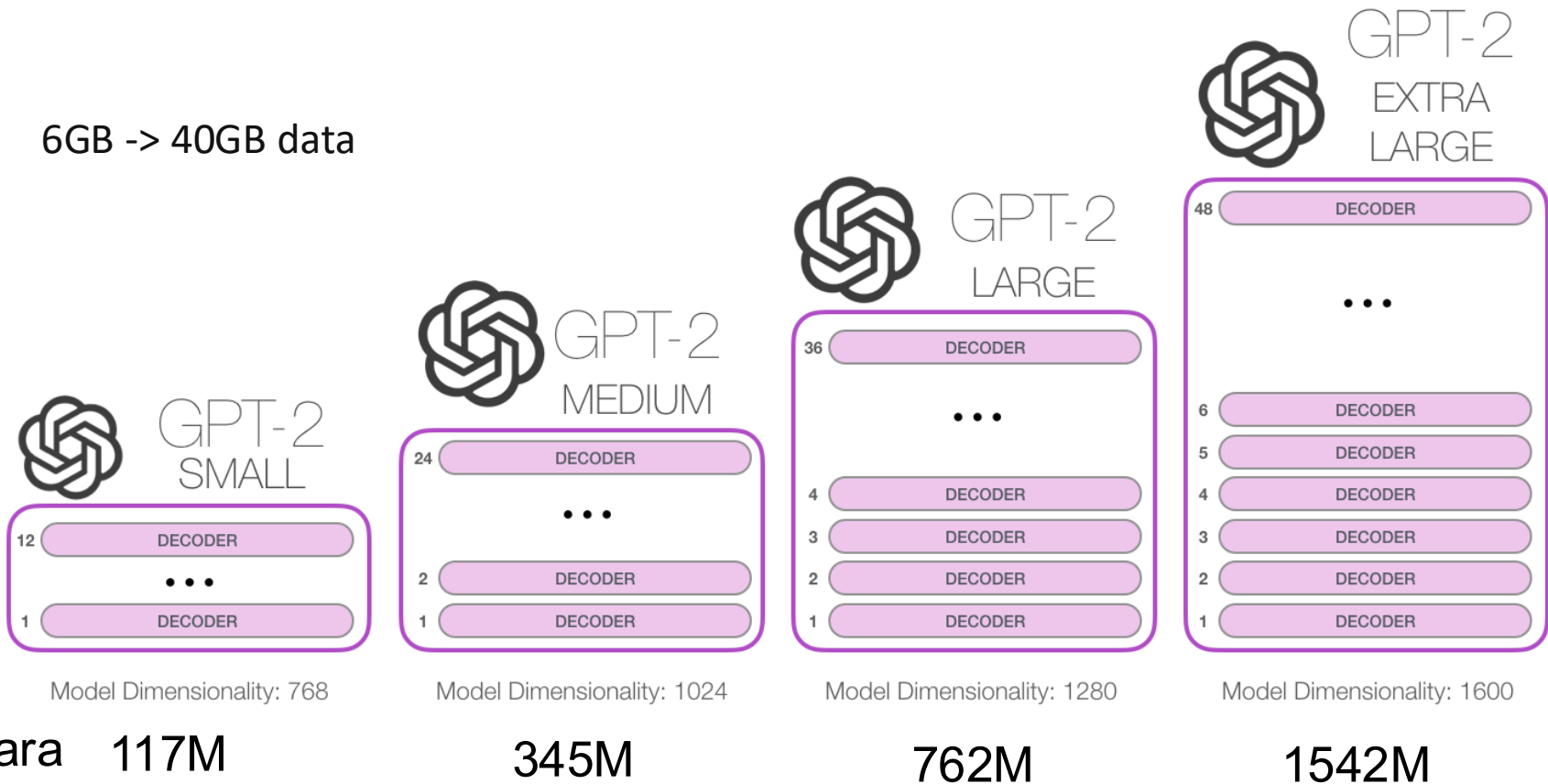
Autoregressive generation



Language Models are unsupervised multitask learners (GPT-2)

The developments in GPT-2 model were mostly in terms of using **a larger dataset** and **adding more parameters** to the model to learn even stronger language model.

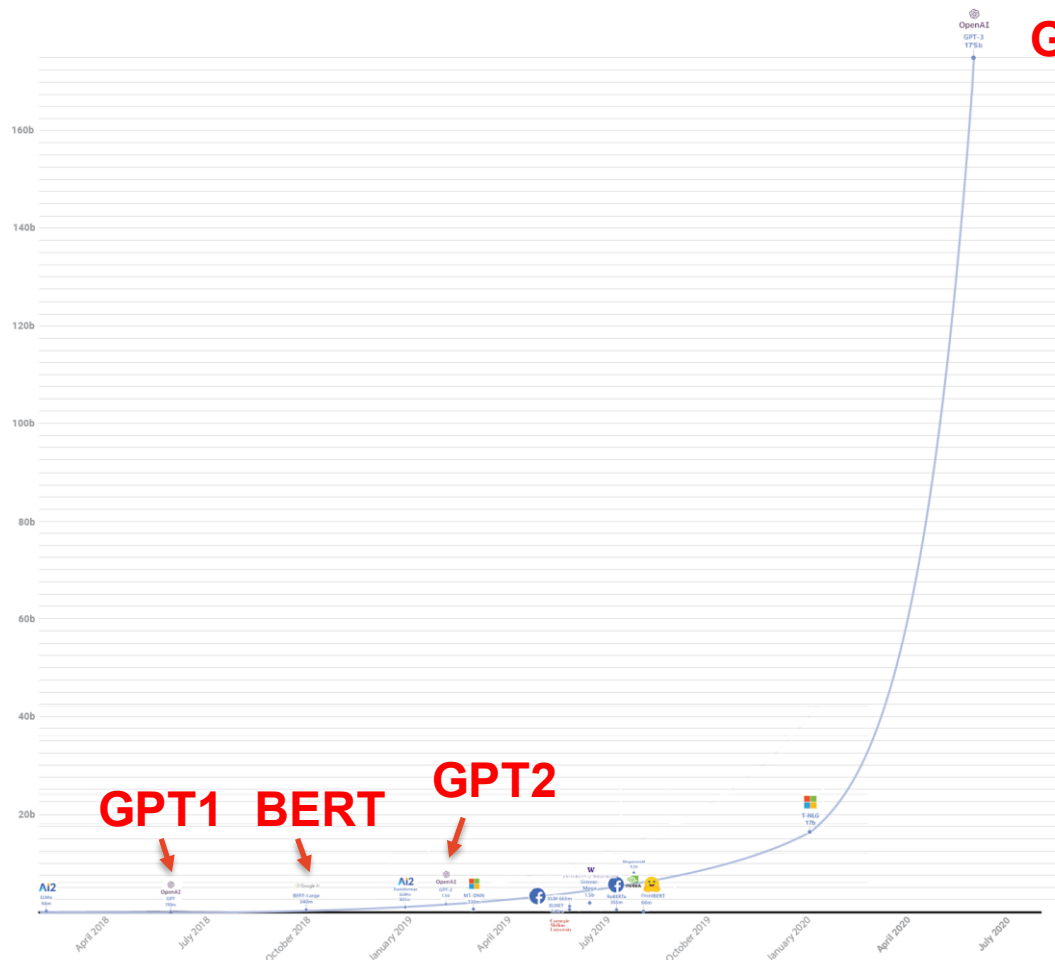
6GB -> 40GB data



Language models are few shot learners (GPT-3):

Same architecture as GPT and GPT-2, vanilla Transformer with small upgrades accumulated over time.

#parameters



GPT3

45TB data
285,000 CPUs
10,000 GPUs
\$12,000,000

GPT-3.5 (InstructGPT)

- Aligning LLMs with user intent by fine-tuning with human feedback
 - Previous LLMs are not aligned with users. That is, they can generate meaningless, untruthful and toxic outputs
- Core Method

Bring **reinforcement learning with human feedback** (RLHF) into the LLM training process.

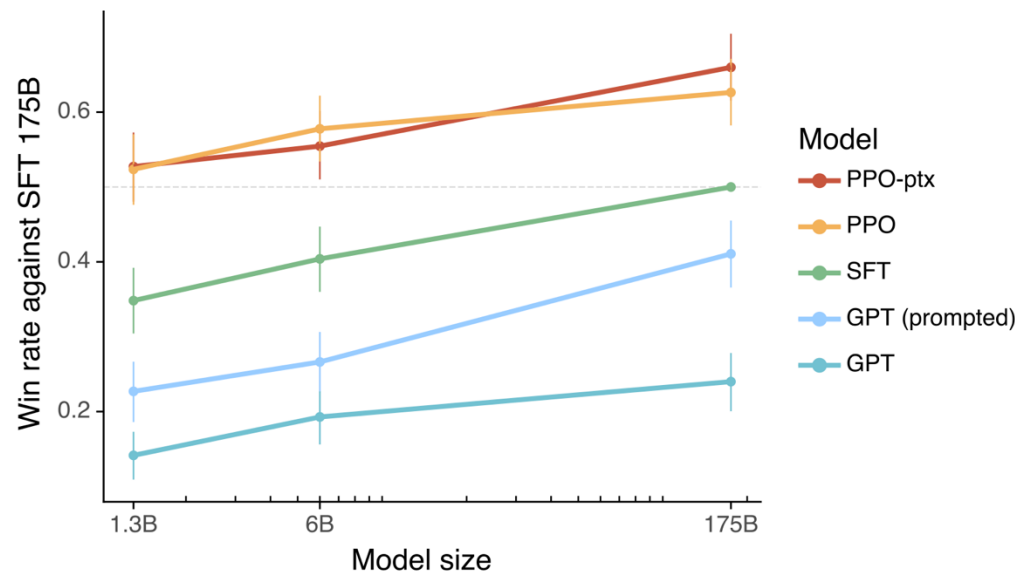
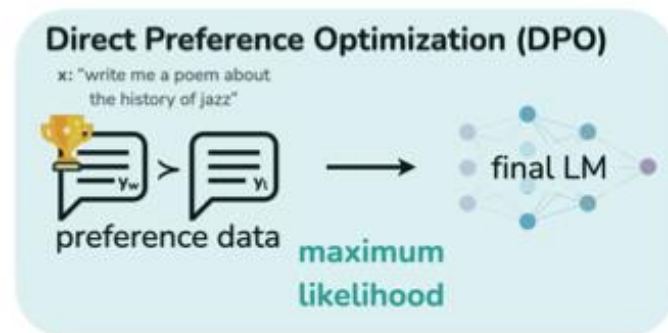
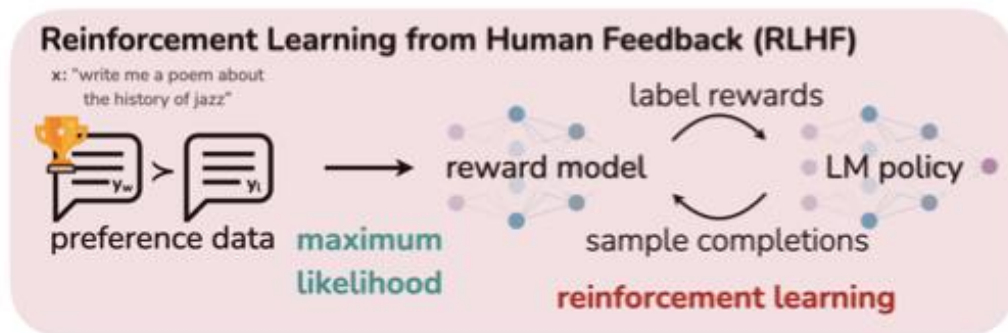


Figure 1: Human evaluations of various models on our API prompt distribution, evaluated by how often outputs from each model were preferred to those from the 175B SFT model. Our InstructGPT models (PPO-ptx) as well as its variant trained without pretraining mix (PPO) significantly outperform the GPT-3 baselines (GPT, GPT prompted); outputs from our 1.3B PPO-ptx model are preferred to those from the 175B GPT-3. Error bars throughout the paper are 95% confidence intervals.

Reinforcement Learning with Human Feedback (RLHF)

A common RLHF method can be divided into 3 stages:

1. Pretraining a Language Model (LM)
2. Gathering data and training a Reward Model (RM)
3. Fine-tuning the LM by reinforcement learning

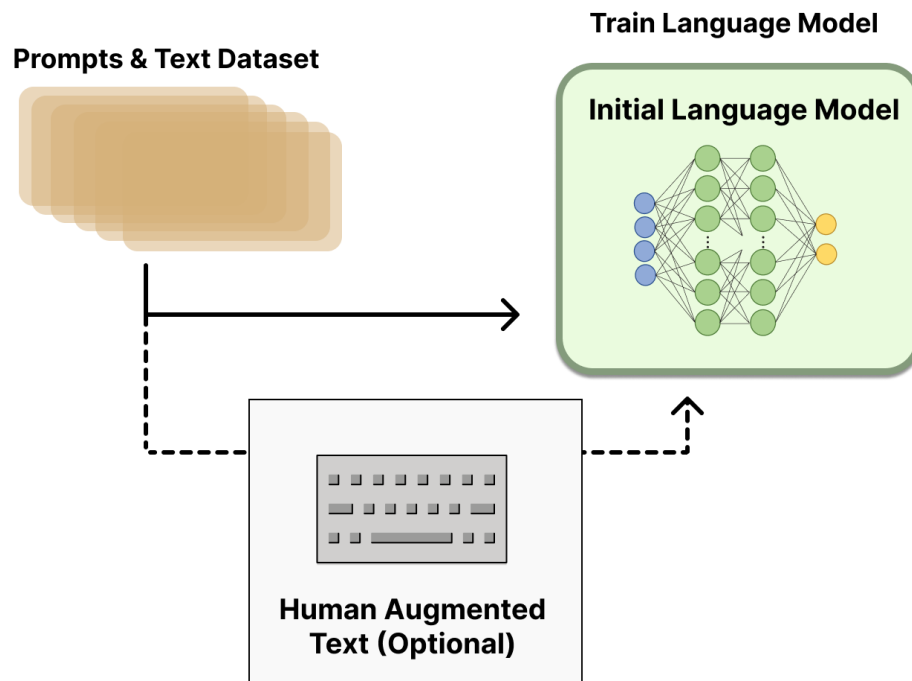


Rafailov, et al., Direct Preference Optimization: Your Language Model is Secretly a Reward Model, NeurIPS 2023

RLHF – Stage 1

Pretraining a Language Model (LM)

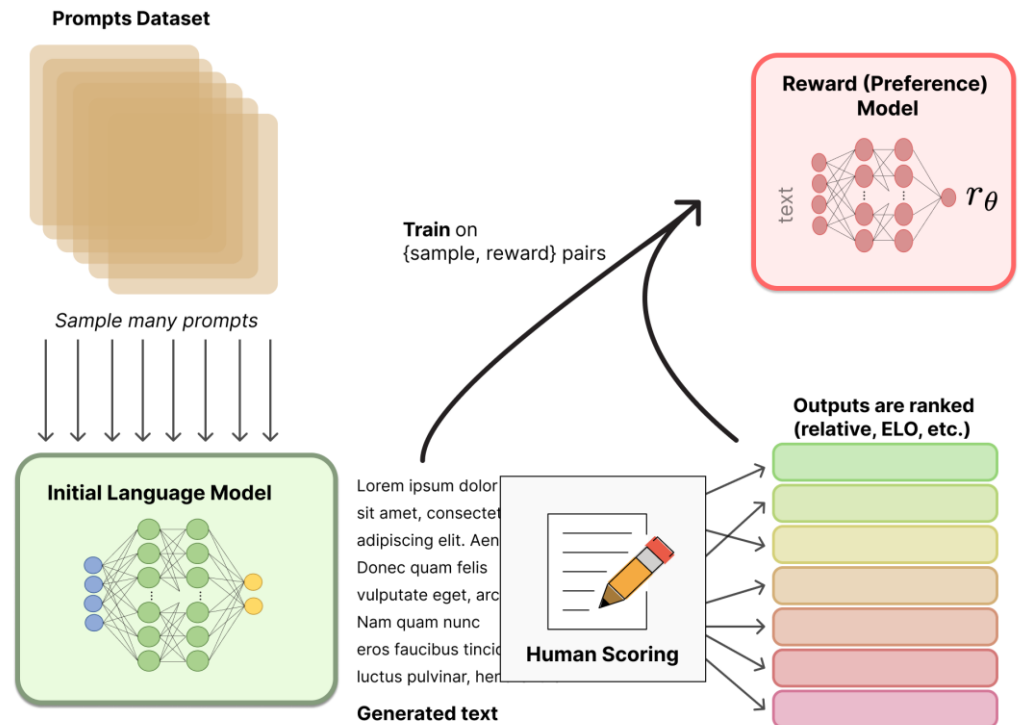
- RLHF generally use pre-trained LMs as the starting point (e.g., GPT-3 is used for GPT-3.5)
- These pre-trained models can be fine-tuned on additional text or conditions, but it is not always necessary. (Human Augmented Text can be used to align human preference)



RLHF – Stage 2

Gathering data and training a Reward Model (RM)

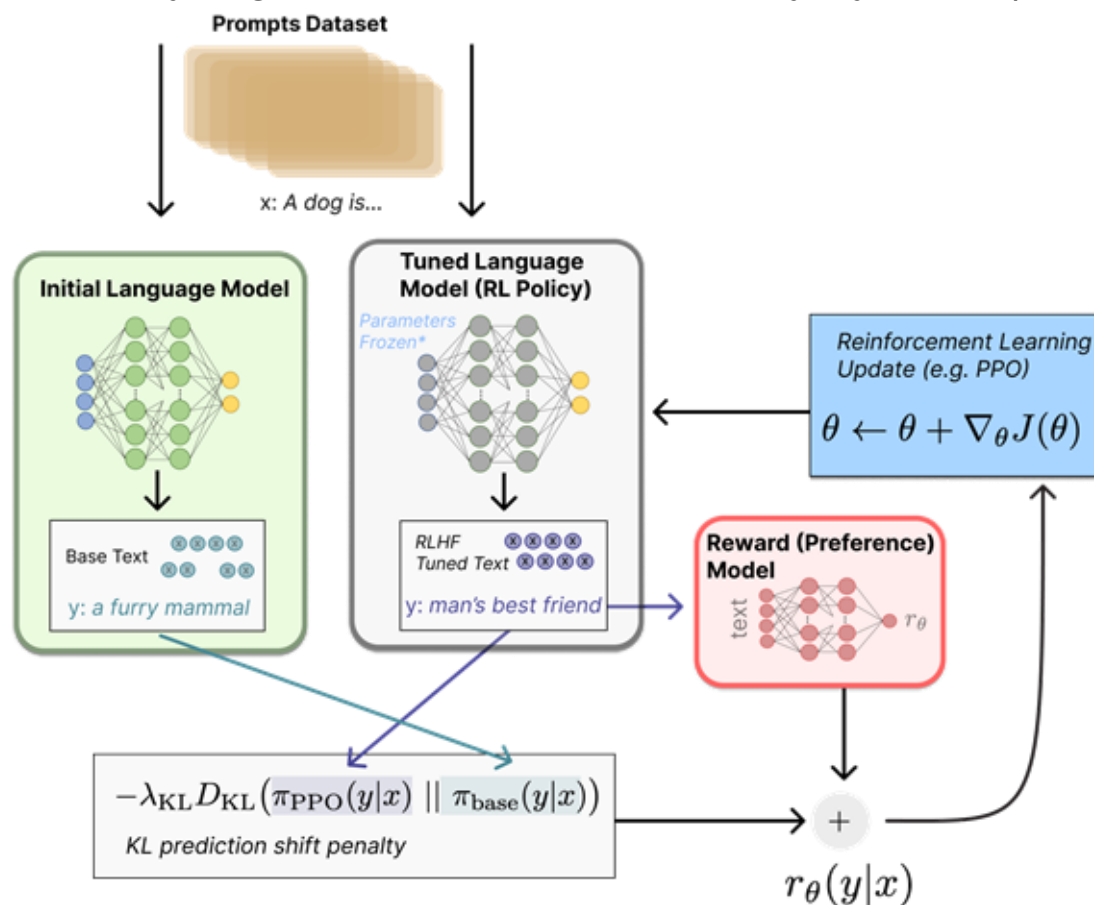
- RM in RLHF: takes in a sequence of text and returns a scalar reward that represent the human preference.
- Training set for the RM:
Generated by sampling prompts and passing them through the initial LM to generate new text, which is then ranked by human annotators.
- Note: Human annotators rank the outputs from the LM rather than scoring them directly.



RLHF – Stage 3

Fine-tuning the LM by reinforcement learning

1. Sample a prompt
2. Pass it through both initial LM and the RL Policy (Copy of the initial LM)
3. Pass the output from policy to RM to calculate reward, and use output from both initial LM and policy to calculate shift penalty
4. Use reward and penalty together to update the policy by PPO (Proximal Policy Optimization)



RLHF in GPT-3.5

Using GPT-3 as starting point

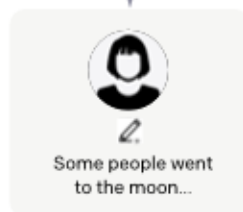
Step 1

Collect demonstration data, and train a supervised policy.

A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3 with supervised learning.



Step 2

Collect comparison data, and train a reward model.

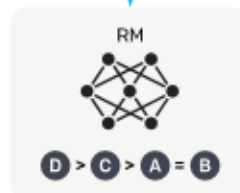
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.



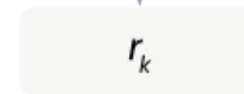
The policy generates an output.



The reward model calculates a reward for the output.

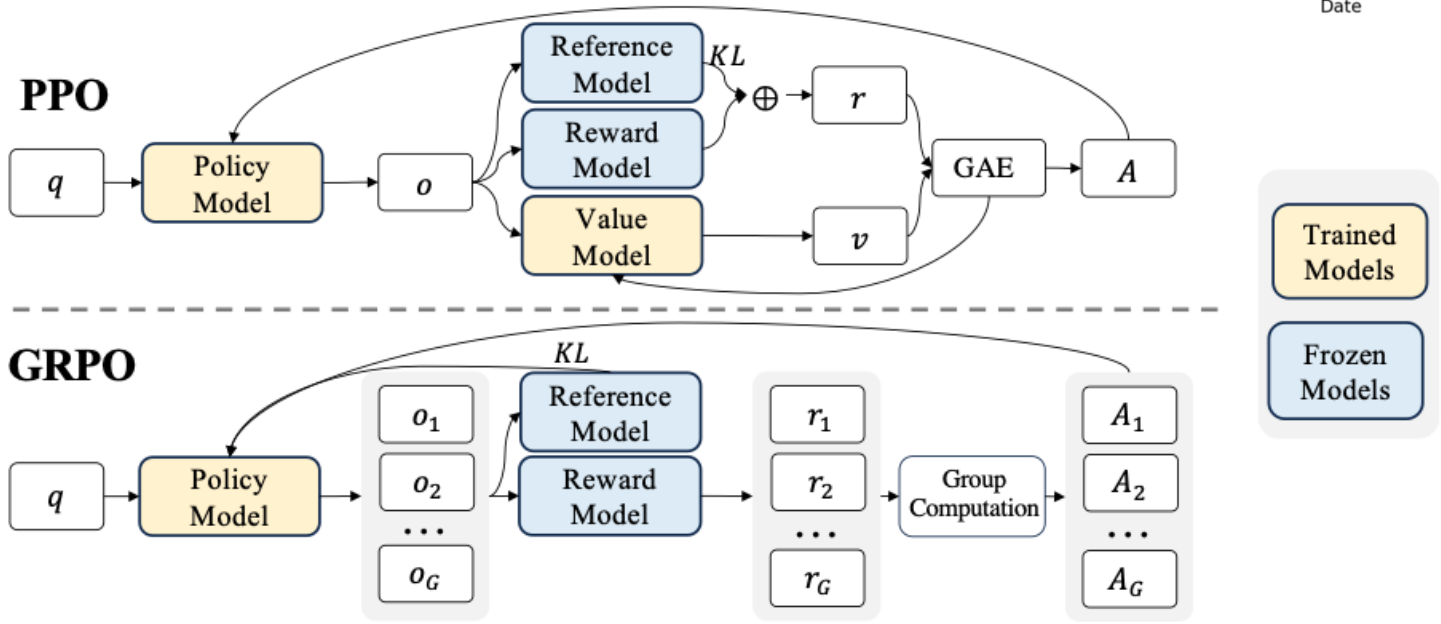
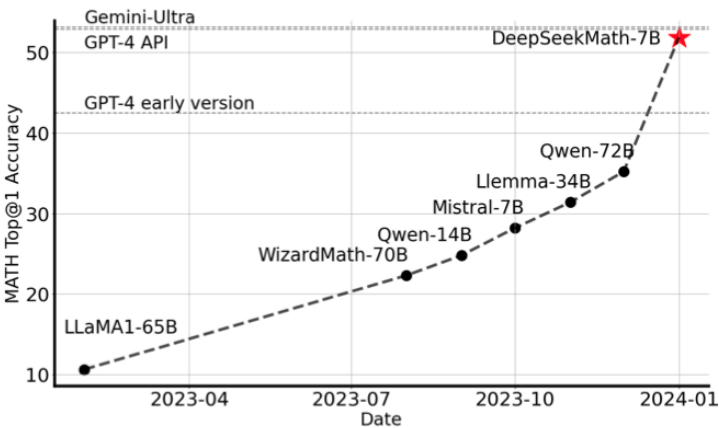


The reward is used to update the policy using PPO.



Group Relative Policy Optimization

Removing the **value model** in PPO



DeepSeek Team, DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. (5000+ citations)

ChatGPT

- Sibling model to InstructGPT, trained to follow an instruction in a prompt and provide a detailed response.
- Same method as InstructGPT, but slightly different data collection setup.
 - For initial supervised fine-tuning data:
 - The trainers acted as both users and AI assistants to create conversation dataset with help of model-written suggestions.
 - Mix the new dialogue dataset with InstructGPT dataset, which is also converted into dialogue format.
 - For reward model data (comparison data):
 - Took conversations that AI trainers had with the chatbot
 - Randomly selected a model-written message, sampled several alternative completions, and had AI trainers rank them.

GPT-4

Comprehensive superiority beyond ChatGPT

- Better Creativity and better reasoning ability
- Multi-modal (text and visual tasks)
- Allow longer context (25,000 words)
 - ChatGPT allow 3,000 words
- Human-level performance in academic and qualification exams
- ...

Methodology is not disclosed. (Mostly based on the method of ChatGPT)



The University of Sydney

User

Can you explain this meme?

Sometimes I just look at pictures of the earth from space and I marvel at how beautiful it all is.



GPT-4

This meme is a joke that combines two unrelated things: pictures of the earth from space and chicken nuggets.

The text of the meme suggests that the image below is a beautiful picture of the earth from space. However, the image is actually of chicken nuggets arranged to vaguely resemble a map of the world.

The humor in this meme comes from the unexpected juxtaposition of the text and the image. The text sets up an expectation of a majestic image of the earth, but the image is actually something mundane and silly.



ChatGPT

The flag in the image represents the "Australian 2/4th Armoured Regiment," and lists battle honours from World War II. The places mentioned — Aitape, Torokina, Rabaul, Slater's Knoll, and Weewak — are locations where this regiment saw action during the war. Such banners are often carried in parades to honor the service and sacrifices of military units.

Emergent Ability

Emergent Abilities refer to the ability of a model to automatically learn and discover new, higher-level features and patterns from the raw training data.

ChatGPT and GPT-4 largely benefit from this.

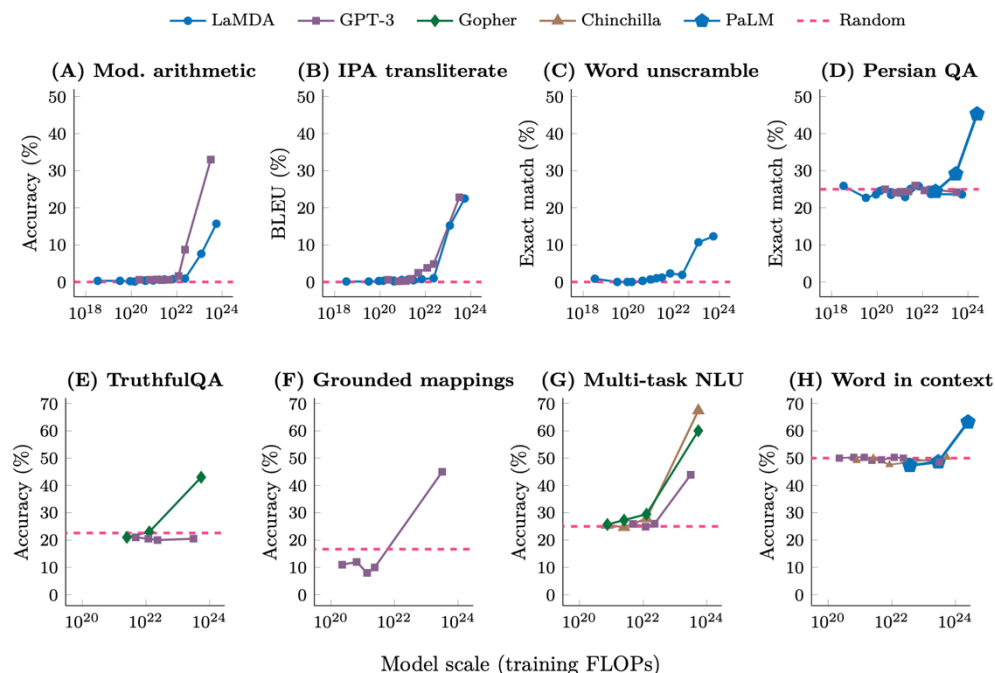
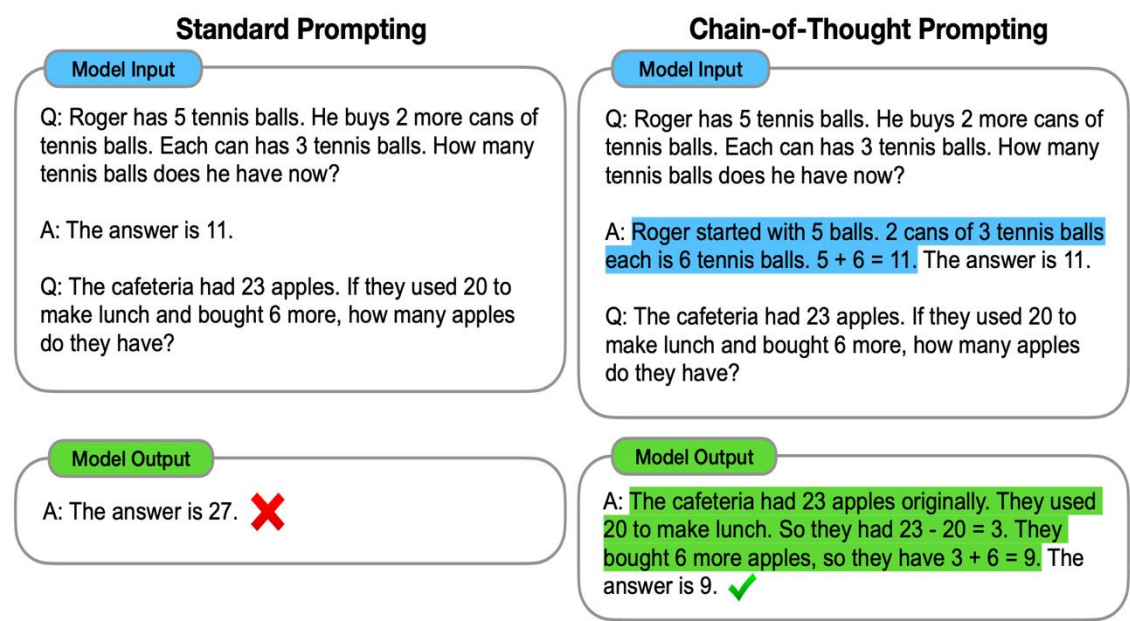


Figure 2: Eight examples of emergence in the few-shot prompting setting. Each point is a separate model. The ability to perform a task via few-shot prompting is emergent when a language model achieves random performance until a certain scale, after which performance significantly increases to well-above random. Note that models that used more training compute also typically have more parameters—hence, we show an analogous figure with number of model parameters instead of training FLOPs as the x -axis in Figure 11. A–D: BIG-Bench (2022), 2-shot. E: Lin et al. (2021) and Rae et al. (2021). F: Patel & Pavlick (2022). G: Hendrycks et al. (2021a), Rae et al. (2021), and Hoffmann et al. (2022). H: Brown et al. (2020), Hoffmann et al. (2022), and Chowdhery et al. (2022) on the WiC benchmark (Pilehvar & Camacho-Collados, 2019).

Chain of Thought

- Generating a Chain of Thought (a series of intermediate reasoning steps) can significantly improve the ability of LLMs to perform complex reasoning
- CoT emerge naturally in sufficiently large language models such as ChatGPT and GPT-4, by adding simple modification in prompt



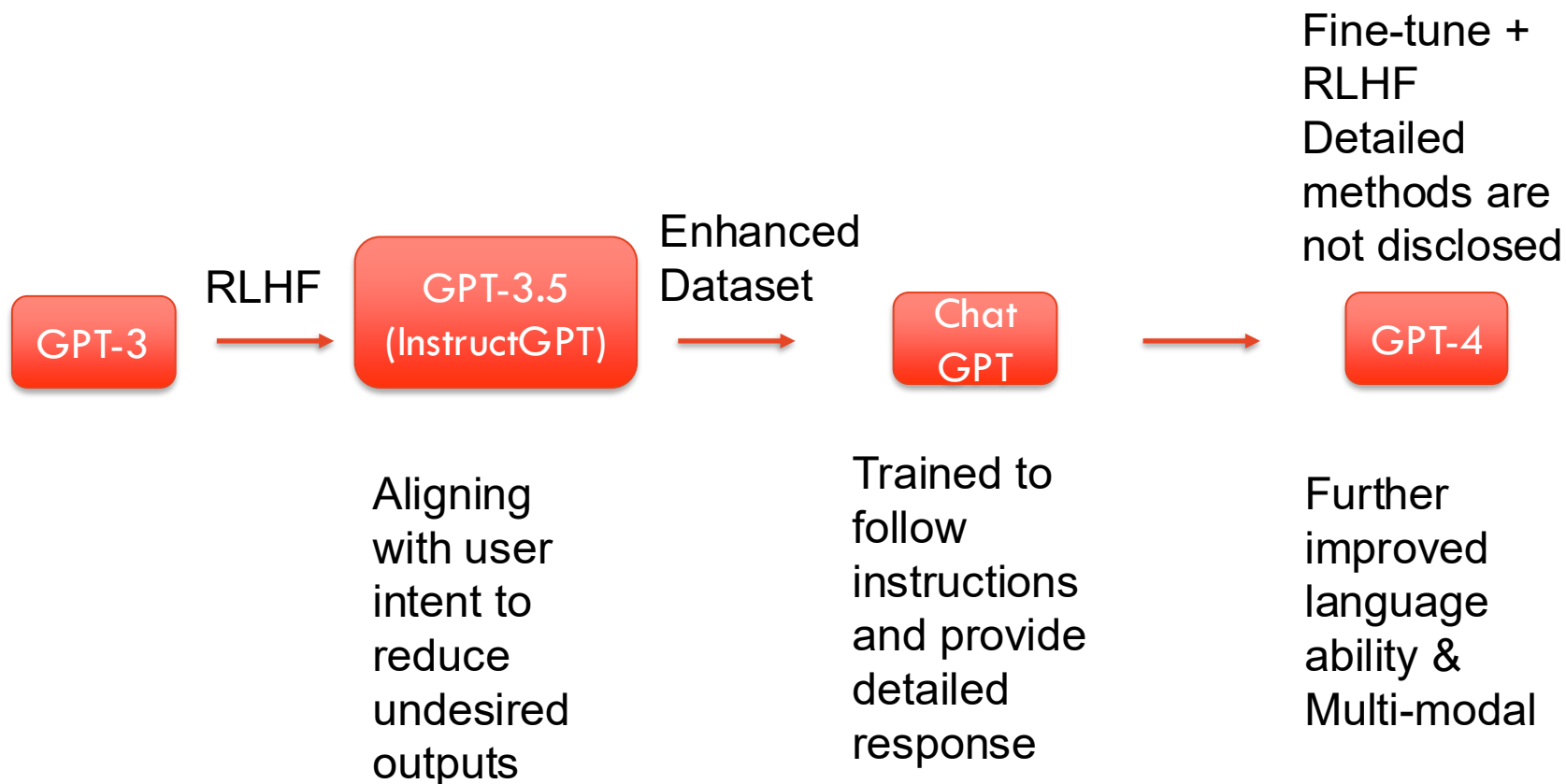
Prompt Engineering

- Prompt is the input instruction of LMs
- Prompt Engineering is about designing and optimizing prompts to further improve the performance of LMs
- The CoT is a typical prompt engineering method, there are also some others:
 - Self-consistency
 - Generated Knowledge prompt
 - Etc.

A good prompt could have following contents:

- Capacity and Role
 - e.g., Now you should act as a teacher
- Context
 - e.g., We are in the deep learning class
- Instruction
 - e.g., Explain the idea of prompt engineering
- Style of response
 - e.g., Your explanation should be easy to understand

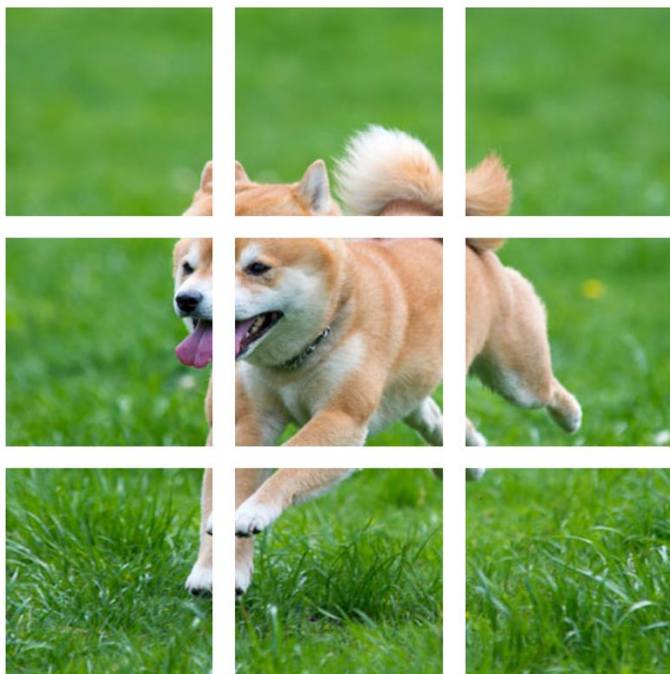
Evolution after GPT-3



Vision Transformer

How to represent an image with a sequence?

1. Split an image into several patches

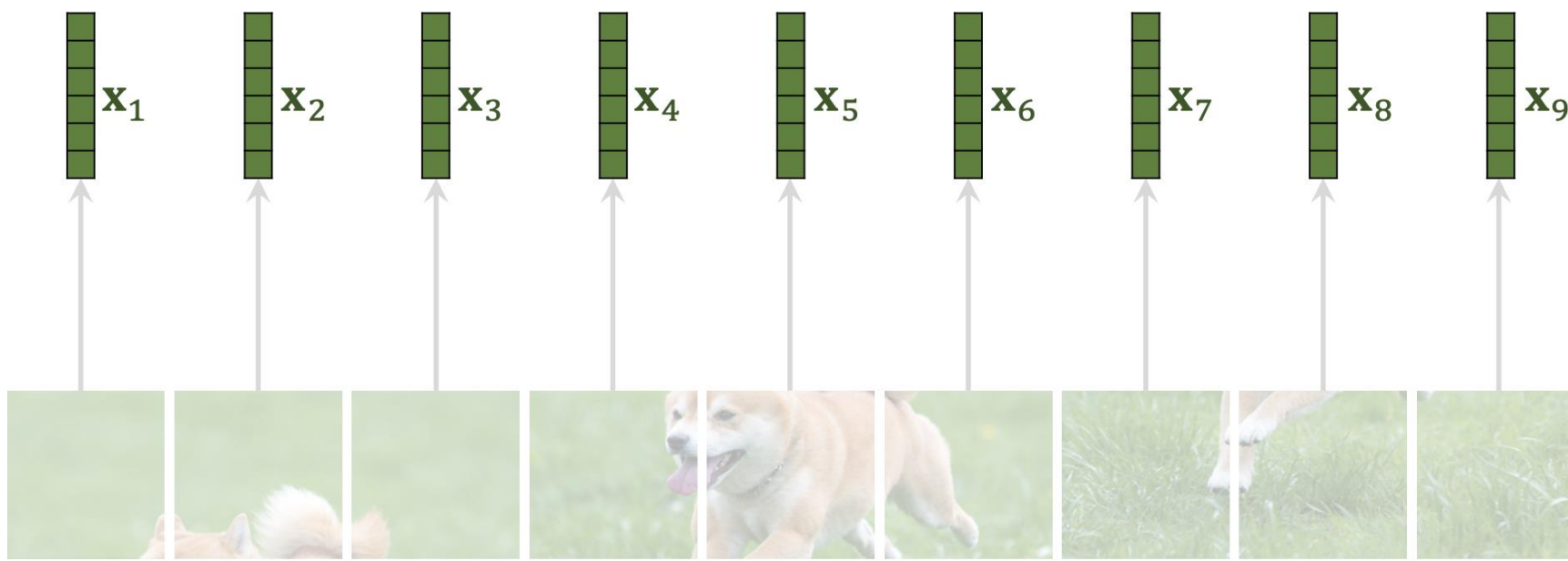


- Here, the patches do not overlap.
- The patches can overlap.
- User specifies:
 - **patch size**, e.g., 16×16 ;
 - **stride**, e.g., 16×16 .

How to represent an image with a sequence?

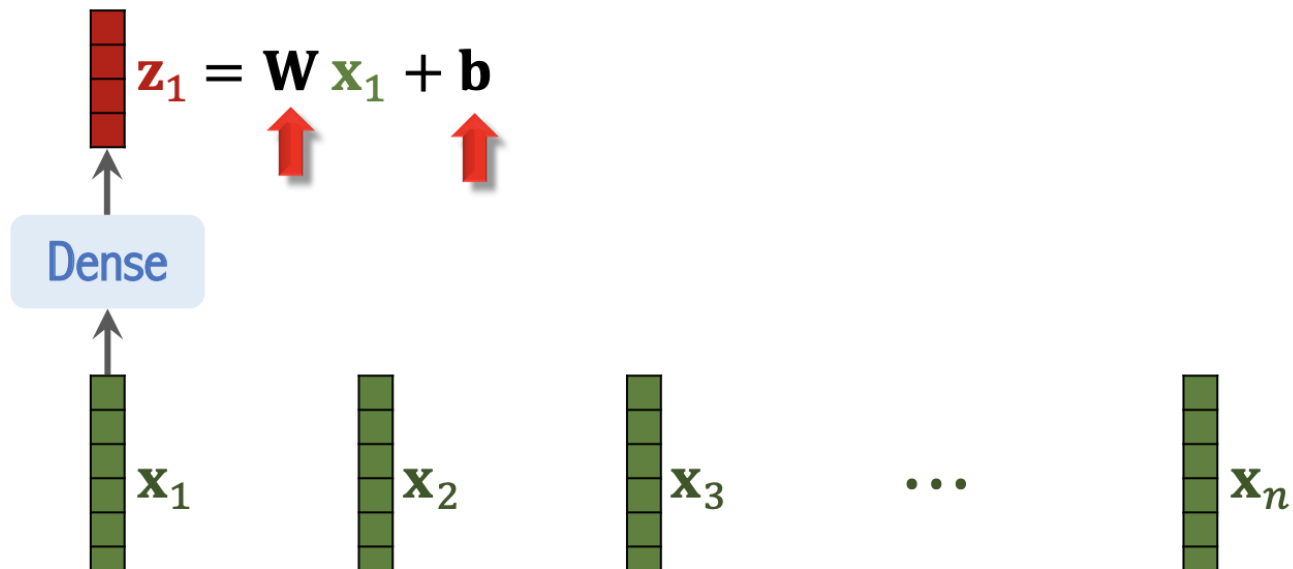
2. Vectorization

If the patches are $d_1 \times d_2 \times d_3$ tensors, then the vectors are $d_1 d_2 d_3 \times 1$.



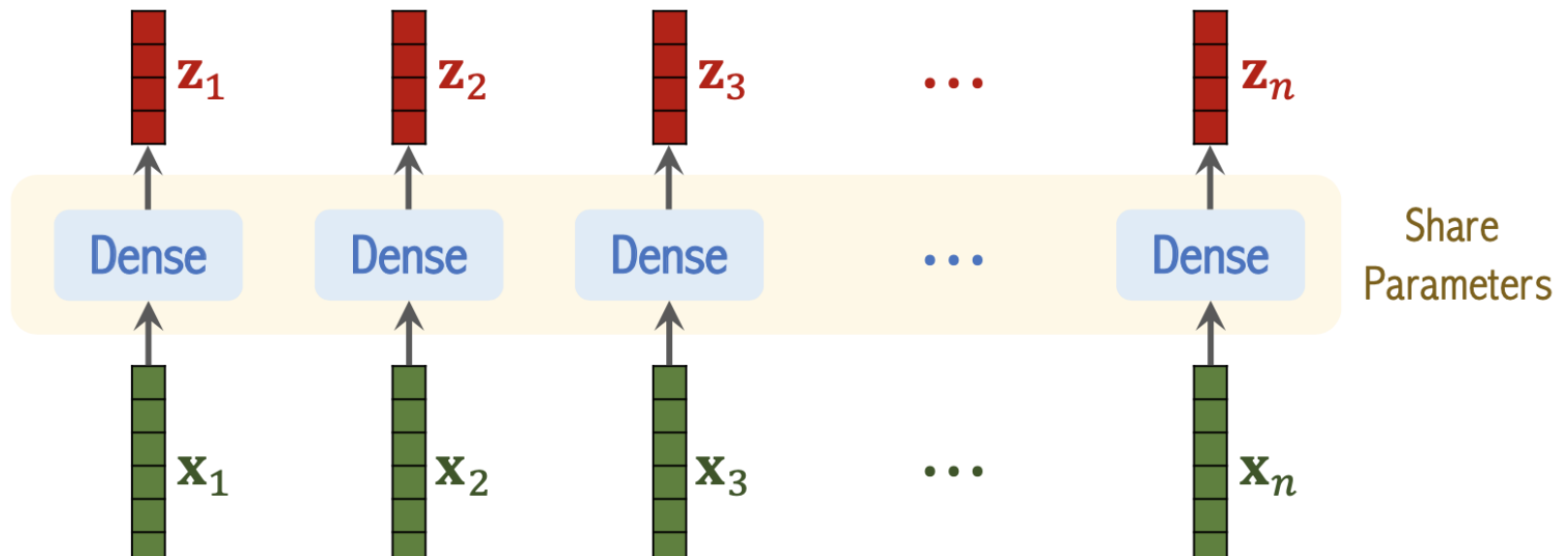
How to represent an image with a sequence?

2. Vectorization



How to represent an image with a sequence?

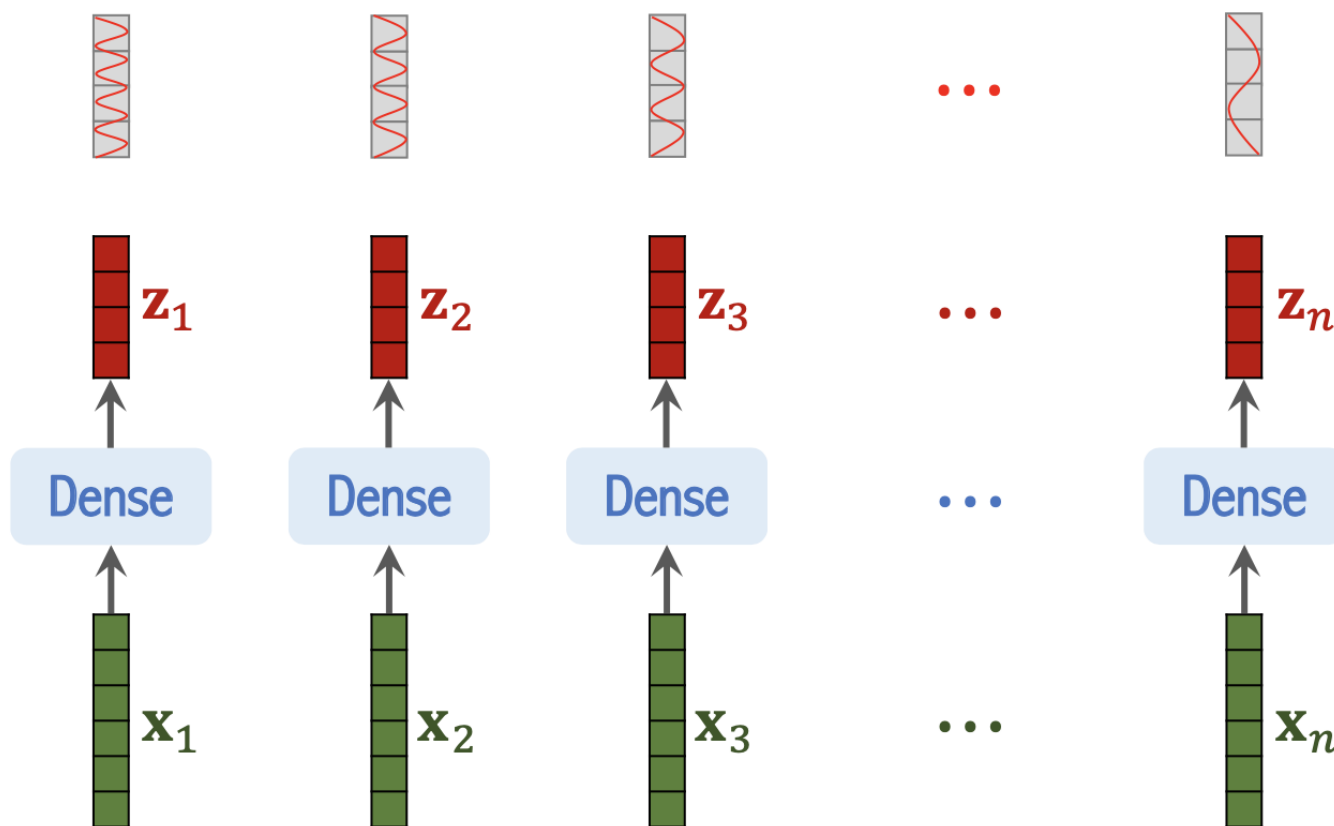
2. Vectorization



How to represent an image with a sequence?

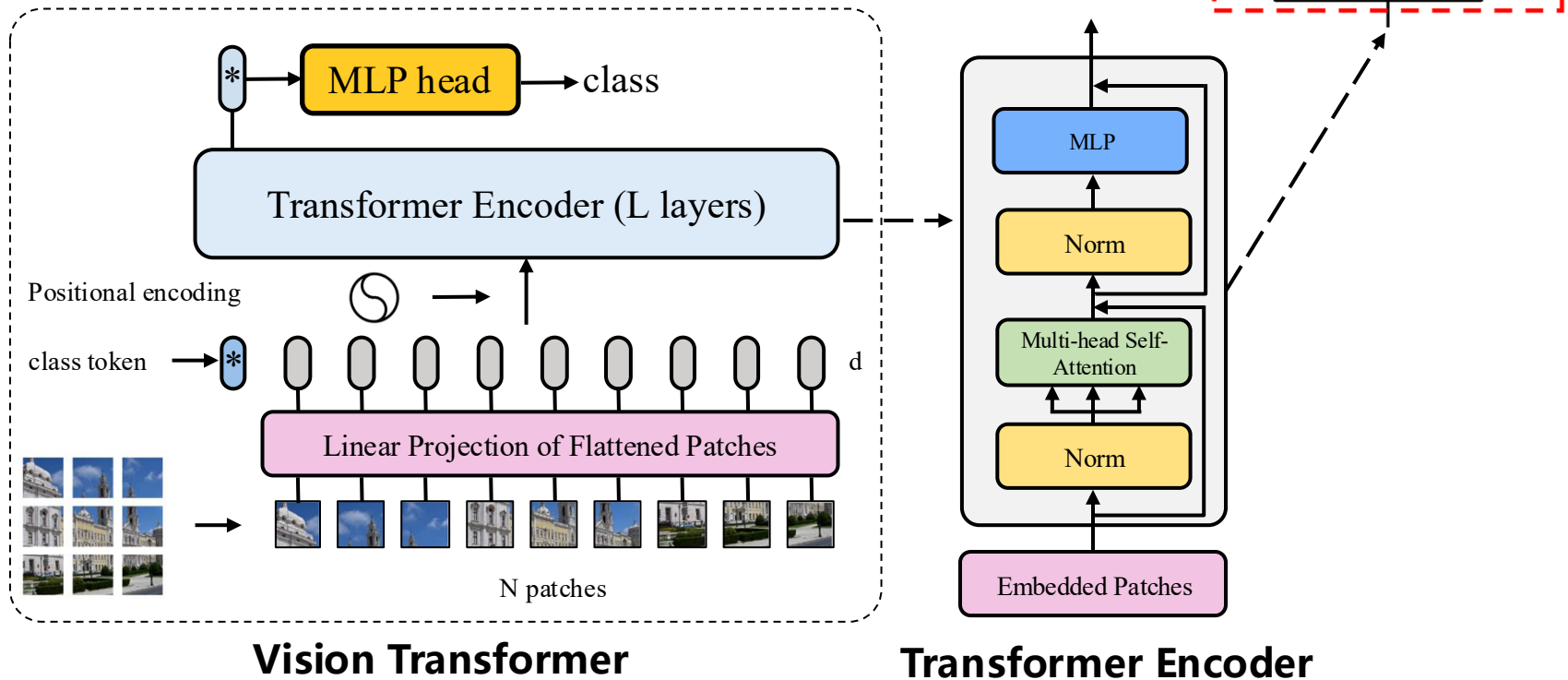
3. Add positional encoding

Add positional encoding vectors to $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n$.



Vision Transformer

4. Overall Framework



Dosovitskiy et al., An Image is Worth 16*16 Words: Transformers for image recognition, ICLR, 2021, Google Brain

Vision Transformer

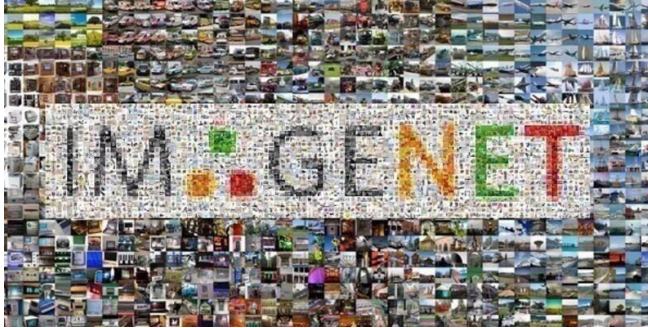
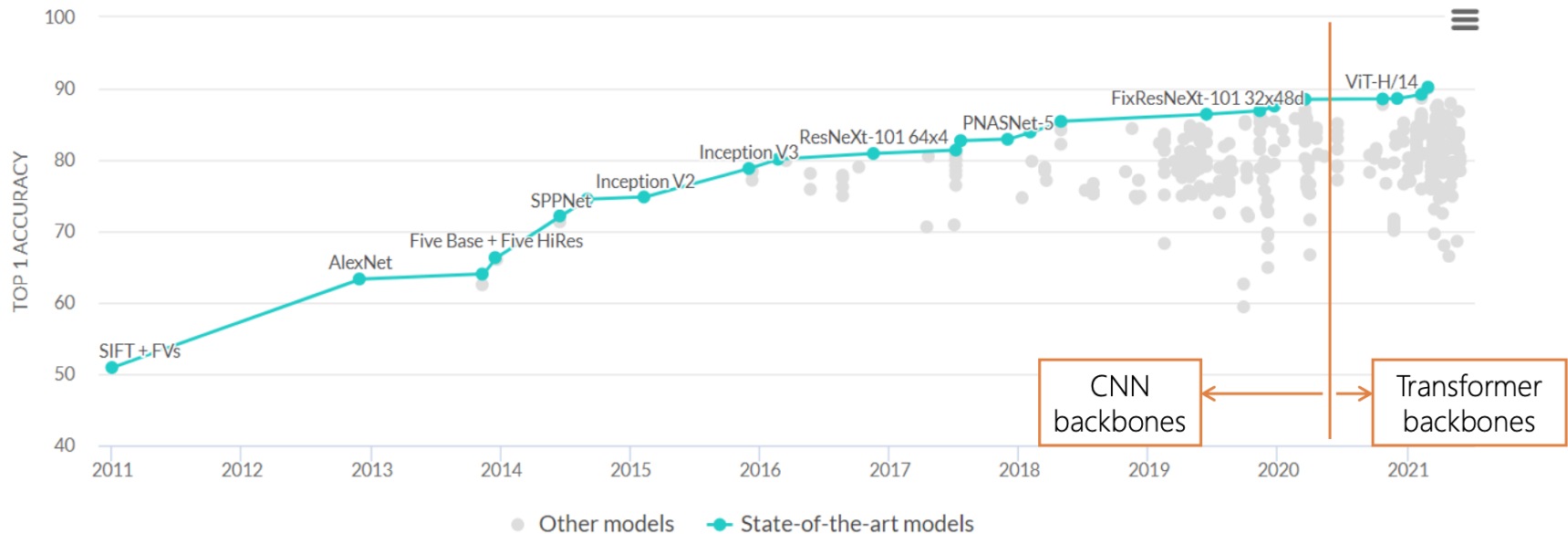


Image Classification Accuracy on ImageNet-1K

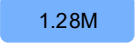
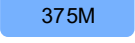
~1.2M Images, 1K Categories



<https://paperswithcode.com/sota/image-classification-on-imagenet>

Vision Transformer

ViT needs a large amount of training data to reach SOTA performance.

		ViT-B/16	ViT-B/32	ViT-L/16	ViT-L/32	ViT-H/14
ImageNet   	CIFAR-10	98.13	97.77	97.86	97.94	-
	CIFAR-100	87.13	86.31	86.35	87.07	-
	ImageNet	77.91	73.38	76.53	71.16	-
	ImageNet ReaL	83.57	79.56	82.19	77.83	-
	Oxford Flowers-102	89.49	85.43	89.66	86.36	-
	Oxford-IIIT-Pets	93.81	92.04	93.64	91.35	-
ImageNet-21k   	CIFAR-10	98.95	98.79	99.16	99.13	99.27
	CIFAR-100	91.67	91.97	93.44	93.04	93.82
	ImageNet	83.97	81.28	85.15	80.99	85.13
	ImageNet ReaL	88.35	86.63	88.40	85.65	88.70
	Oxford Flowers-102	99.38	99.11	99.61	99.19	99.51
	Oxford-IIIT-Pets	94.43	93.02	94.73	93.09	94.82
JFT-300M   	CIFAR-10	99.00	98.61	99.38	99.19	99.50
	CIFAR-100	91.87	90.49	94.04	92.52	94.55
	ImageNet	84.15	80.73	87.12	84.37	88.04
	ImageNet ReaL	88.85	86.27	89.99	88.28	90.33
	Oxford Flowers-102	99.56	99.27	99.56	99.45	99.68
	Oxford-IIIT-Pets	95.80	93.40	97.11	95.83	97.56

EfficientNet-B7
Top-1 Acc. 84.3%

Vision Transformer

Improved Variants

🔗 Swin Transformer

🏆 Ranked #3 Object Detection on COCO test-dev (using additional training data)

🏆 Ranked #2 Instance Segmentation on COCO test-dev (using additional training data)

🏆 Ranked #2 Object Detection on COCO minival (using additional training data)

🏆 State of the Art Instance Segmentation on COCO minival (using additional training data)

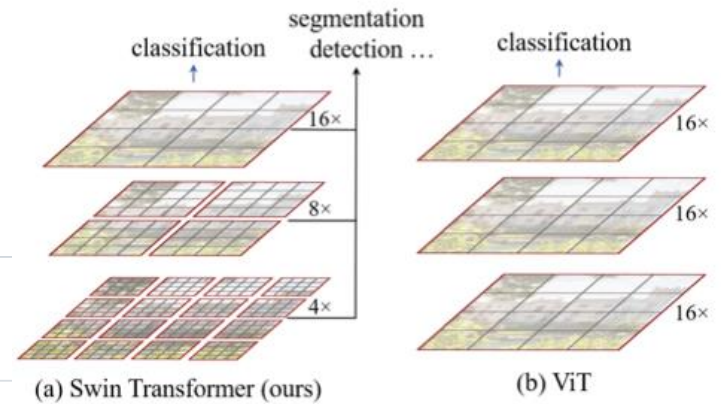
🏆 Ranked #11 Semantic Segmentation on ADE20K (using additional training data)

🏆 Ranked #11 Semantic Segmentation on ADE20K val

🏆 State of the Art Action Recognition on Something-Something V2 (using additional training data)

🏆 Ranked #3 Action Classification on Kinetics-400 (using additional training data)

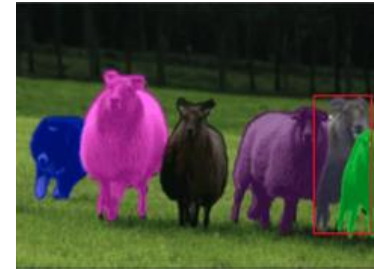
🏆 Ranked #2 Action Classification on Kinetics-600 (using additional training data)



Object Detection



Semantic Segmentation



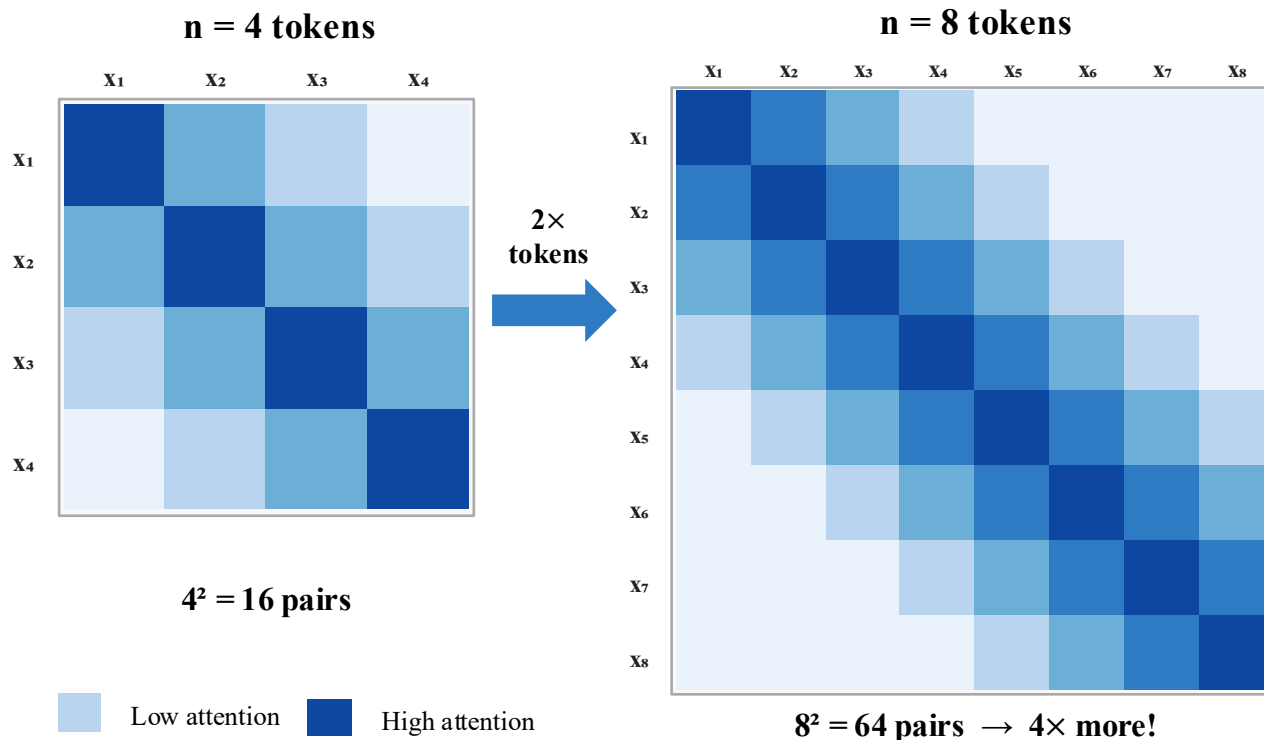
Instance Segmentation

Liu et al., Swin Transformer: Hierarchical Vision Transformer using Shifted Windows, ICCV 2021 (**Best Paper**).

Mamba – A new S2S Architecture

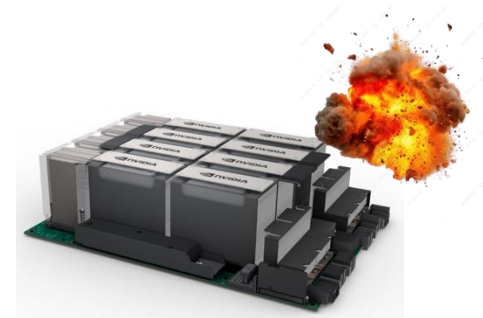
Why Transformer Struggles with Long Sequences?

- ❑ Self-attention computes a score between every pair of tokens:
cost grows as $O(n^2)$
- ❑ Doubling the sequence length quadruples the attention matrix — **memory & compute explode**

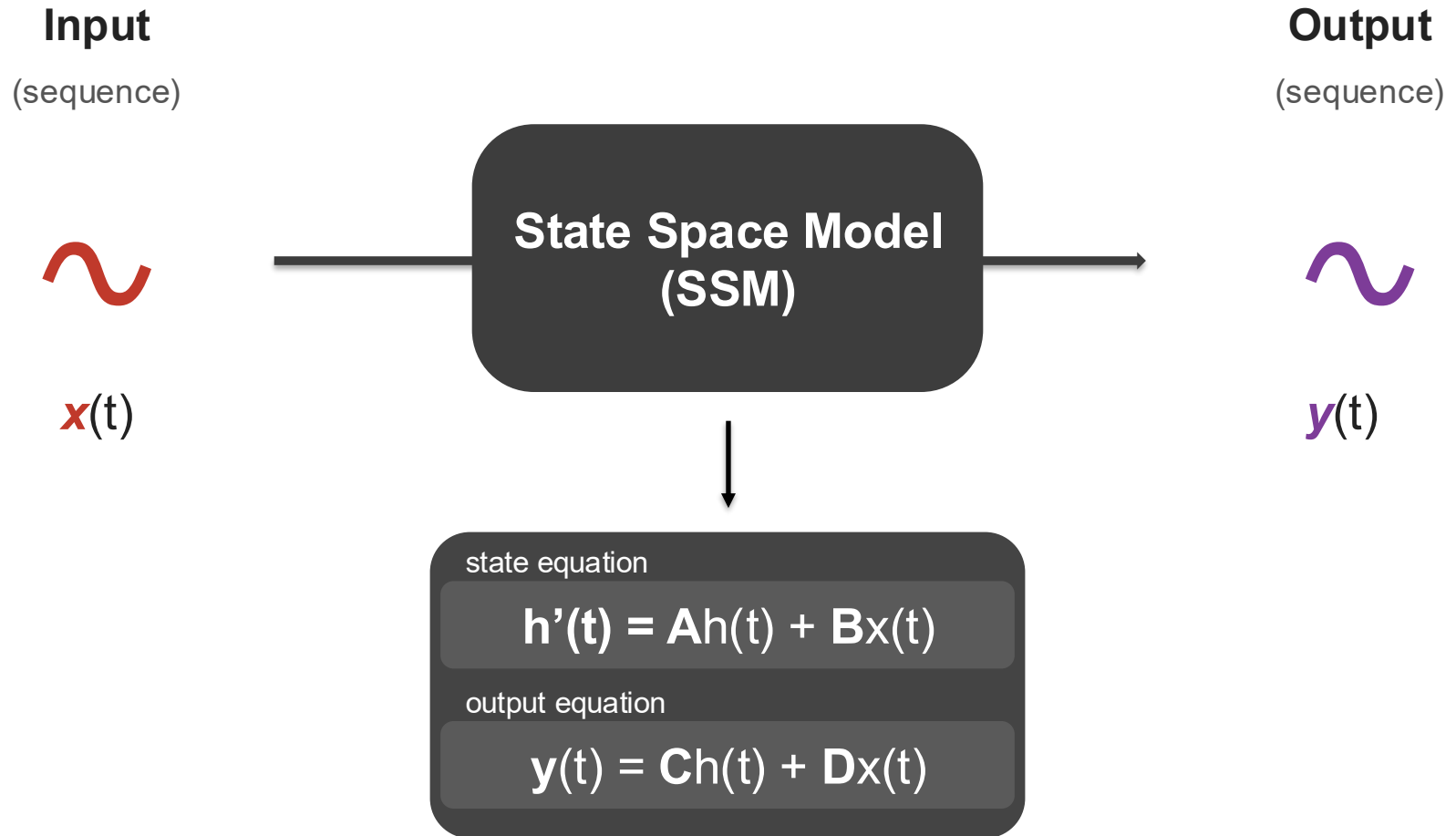


Context Window :

- GPT-3: 4K
- GPT-4: 32K
- GPT-5: ~200K

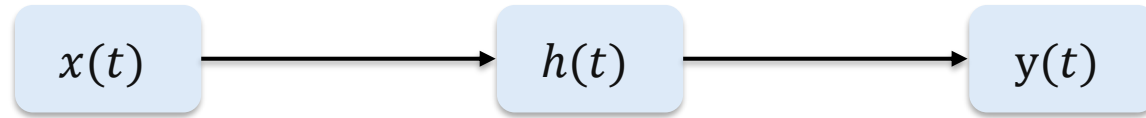


State Space Models



R. E. Kalman. "A New Approach to Linear Filtering and Prediction Problems." ASME 1960

Computing with SSM: Recurrent View



$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t)$$

$$y(t) = \mathbf{C}h(t)$$

Euler Discretization:

$$h(t + \Delta) \approx h(t) + \Delta h'(t) \quad (\text{First-order approximation})$$

$$= h(t) + \Delta(\mathbf{A}h(t) + \mathbf{B}x(t))$$

$$= (\mathbf{I} + \Delta\mathbf{A})h(t) + \Delta\mathbf{B}x(t)$$

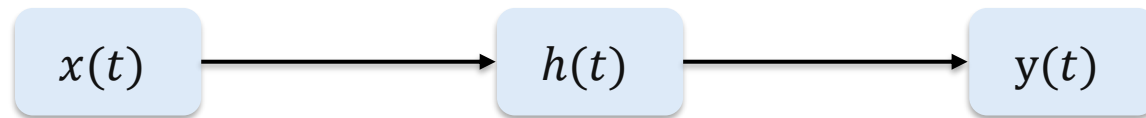
$$= \bar{\mathbf{A}}h(t) + \bar{\mathbf{B}}x(t)$$

h_{t+1}

h_t

x_t

Computing with SSM: Recurrent View



$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t)$$

$$y(t) = \mathbf{C}h(t)$$

1. Discretize

$$\bar{\mathbf{A}} = (\mathbf{I} + \Delta\mathbf{A}) \quad \bar{\mathbf{B}} = \Delta\mathbf{B}$$

2. Recurrent “hidden state”

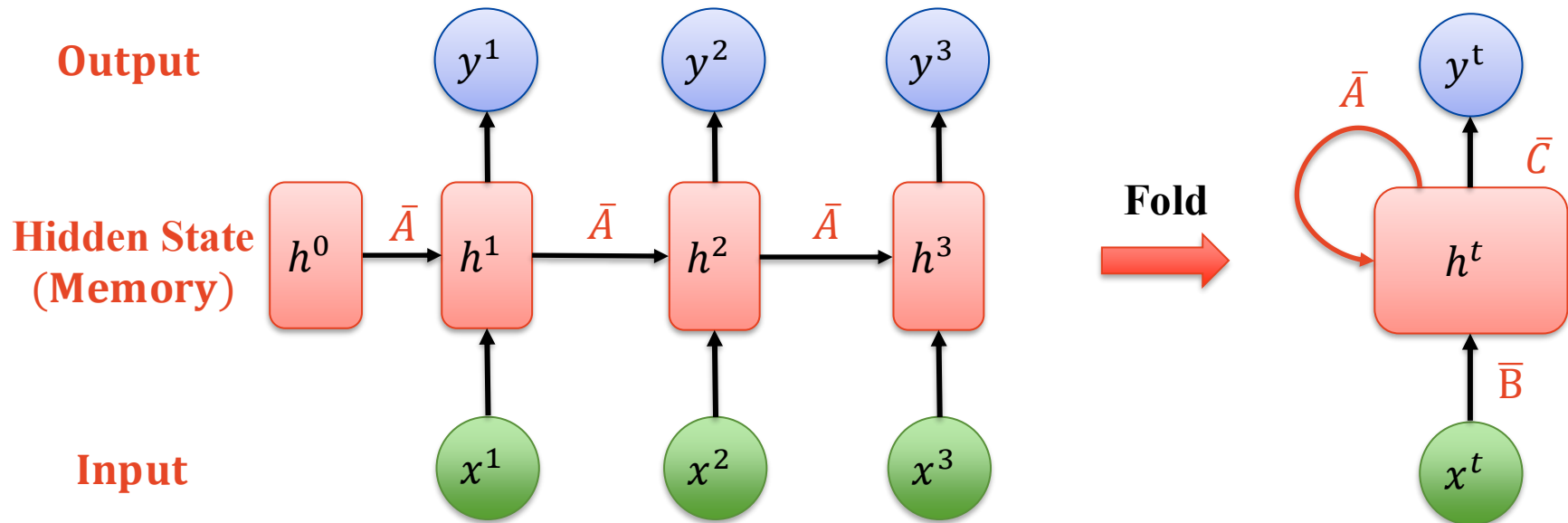
$$h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t$$

3. Out projection

$$y_t = \bar{\mathbf{C}}h_t$$

Can be unrolled as a linear RNN

Recurrent Computation

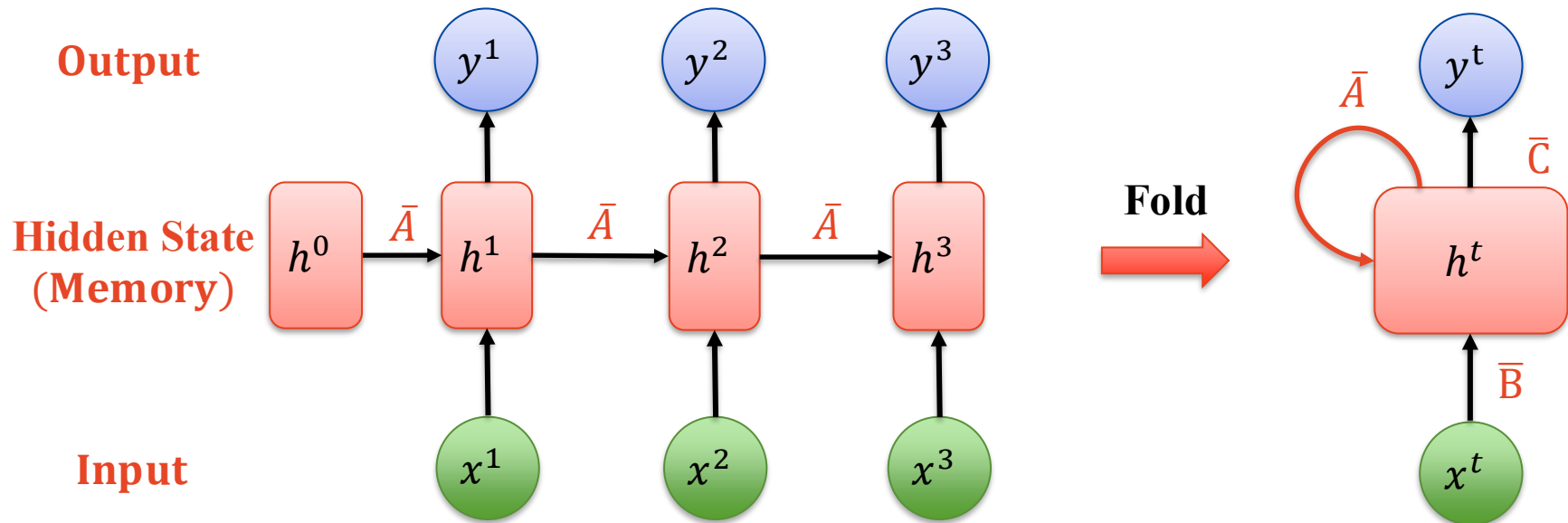


- Infinite Context Window
- Complexity: $O(N)$
- Not **parallelizable?**

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t$$

$$y_t = \bar{C}h_t$$

Recurrent Computation



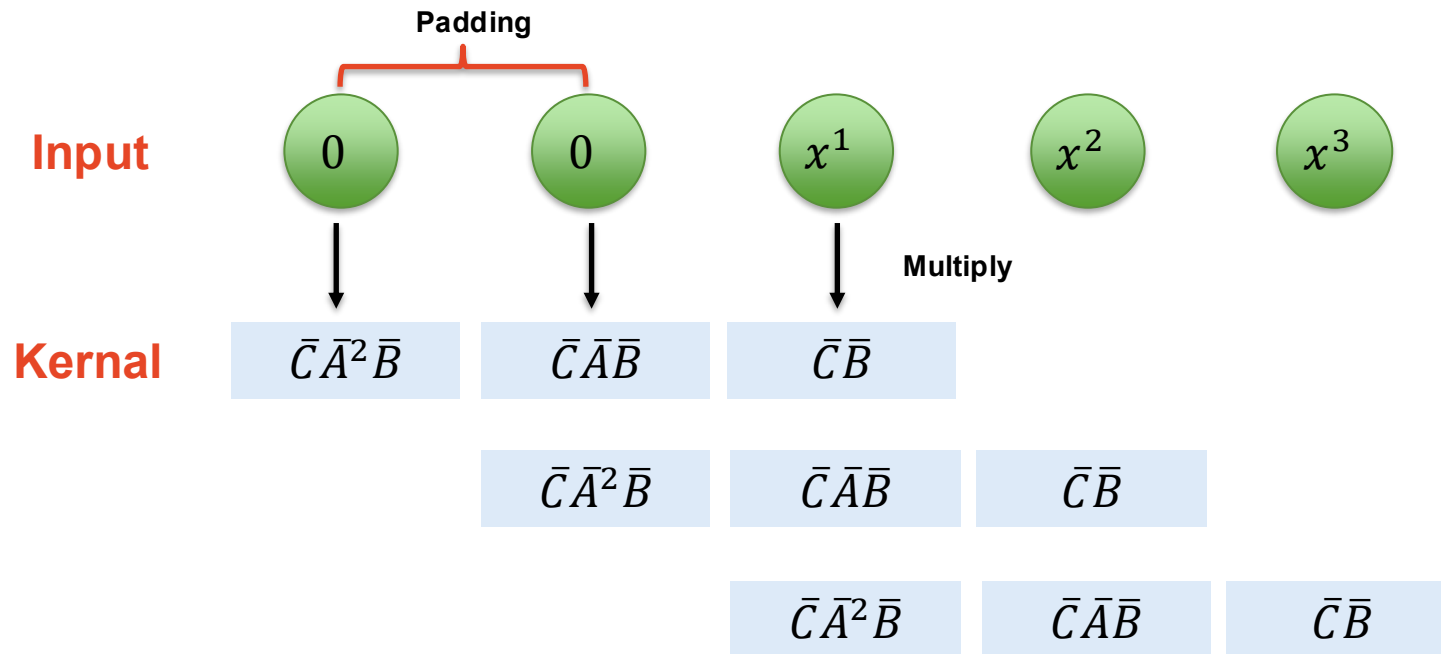
$$h_1 = \bar{A}h_0 + \bar{B}x_1$$

$$h_2 = \bar{A}h_1 + \bar{B}x_2 = \bar{A}(\bar{A}h_0 + \bar{B}x_1) + \bar{B}x_2 = \bar{A}^2h_0 + \bar{A}\bar{B}x_1 + \bar{B}x_2$$

⋮

$$h_t = \bar{A}^t h_0 + \sum_{t=1}^L \bar{A}^{L-t} \bar{B} x_t \quad \longrightarrow \quad y_t = \sum_{t=0}^L \bar{C} \bar{A}^{L-t} \bar{B} x_t$$

Represent SSMs with Convolution



Output

$$y^1 = \bar{C}\bar{B}x^1$$

$$y^2 = \bar{C}\bar{A}\bar{B}x^1 + \bar{C}\bar{B}x^2$$

$$y^3 = \bar{C}\bar{A}^2\bar{B}x^1 + \bar{C}\bar{A}\bar{B}x^2 + \bar{C}\bar{B}x^3$$

$$K = (\bar{C}\bar{A}^2\bar{B}, \bar{C}\bar{A}\bar{B}, \bar{C}\bar{B})$$

$$Y = K * X$$

$$y^1$$

$$y^2$$

$$y^3$$

Training and Inference of SSMs

Training Mode

$$K \in \mathbb{R}^L = (\bar{C}\bar{A}^{L-1}\bar{B}, \bar{C}\bar{A}^{L-2}\bar{B}, \dots, \bar{C}\bar{A}\bar{B}, \bar{C}\bar{B})$$

➤ Complexity: $O(n \cdot \log n)$

$$Y = K * X$$

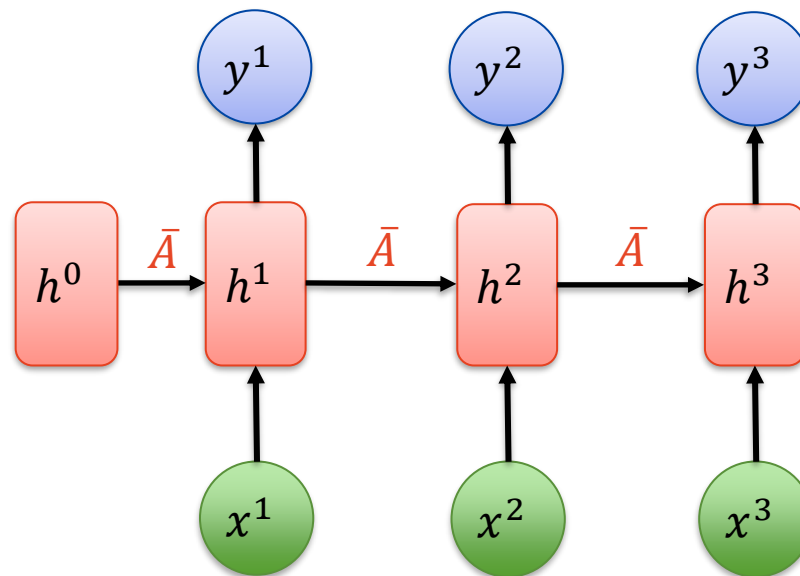
➤ Parallelizable

Can be computed with convolutions, similar to CNN

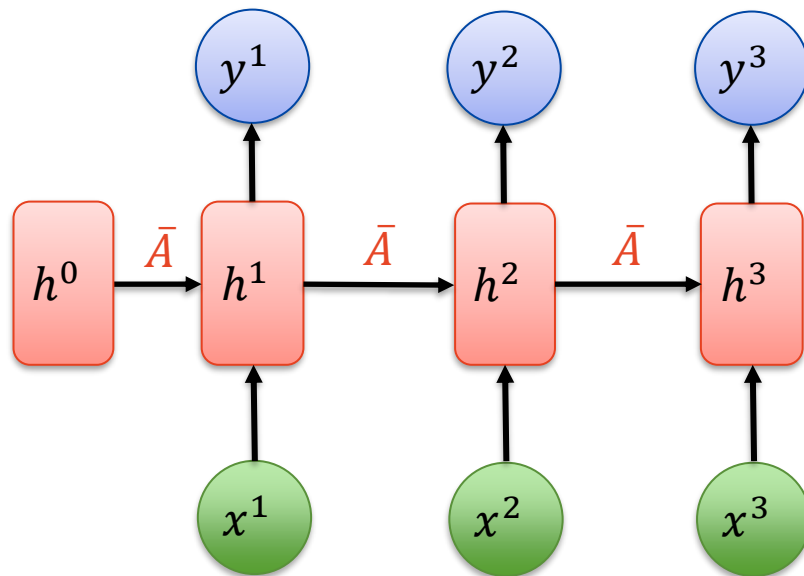
Inference Mode

➤ Infinite Context

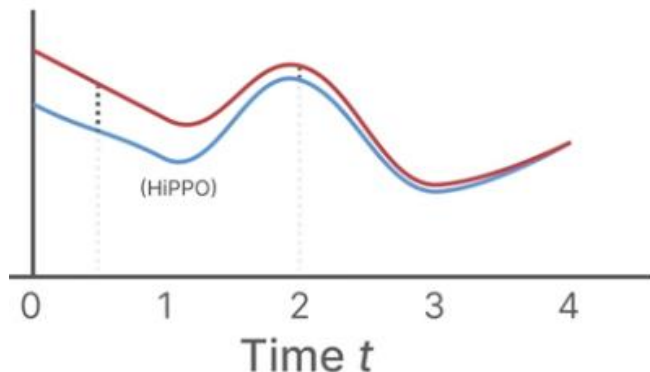
➤ Complexity: $O(1)$



The Importance of A Matrix



The **A** matrix captures information about the previous state to build the new state.



HiPPO

$$A_{nk} = \begin{cases} 0 & n < k \\ n + 1 & n = k \\ 2n + 1 & n > k \end{cases}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 3 & 3 & 0 \\ 1 & 3 & 5 & 4 \end{bmatrix}$$

With Hippo, the model does not forget something at the beginning of a sequence.

Gu et al., HiPPO: Recurrent Memory with Optimal Polynomial Projections, NeurIPS 2020.

Structured State Space Models (S4)

Initialize Parameters

A (with Hippo), B, C

Computer Kernal

$K =$

$(\bar{C}\bar{A}^{L-1}\bar{B}, \bar{C}\bar{A}^{L-2}\bar{B}, \dots, \bar{C}\bar{A}\bar{B}, \bar{C}\bar{B})$

Convolution

$Y = X * K$

```
def parameters(N, dt_min=1e-3, dt_max=1e-1):
    # Initialization
    log_dt = np.random.rand() * (np.log(dt_max)-np.log(dt_min)) + np.log(dt_min)
    A = -0.5 + 1j * np.pi * np.arange(N) # S4D-Lin initialization
    B = np.ones(N)
    C = np.random.randn(N) + 1j * np.random.randn(N) # Variance preserving initia
    return log_dt, np.log(-A.real), A.imag, B, C

def kernel(L, log_dt, log_A_real, A_imag, B, C):
    # Discretization (e.g. bilinear transform)
    dt, A = np.exp(log_dt), -np.exp(log_a_real) + 1j * a_imag
    dA, dB = (1+dt*A/2) / (1-dt*A/2), dt*B / (1-dt*A/2)

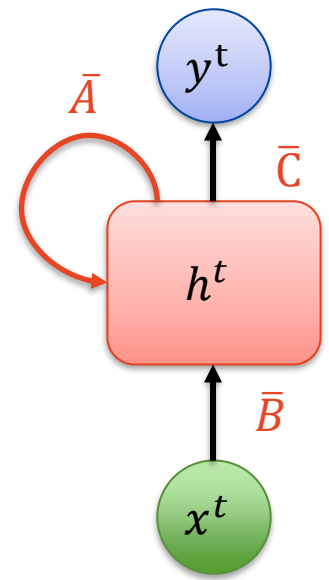
    # Computation (Vandermonde matrix multiplication - can be optimized)
    return (B*C) @ (dA[:, None] ** np.arange(L))

def forward(u, parameters):
    L = u.shape[-1]
    K = kernel(L, *parameters)
    K_f, u_f = np.fft.fft(K, n=2*L), np.fft.fft(input, n=2*L) # Convolve y = u
    return np.fft.ifft(K_f*u_f, n=2*L)[..., :L]
```

Structured State Space Models (S4)

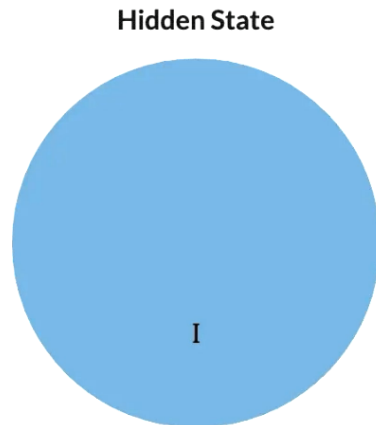
Limitation of S4

- B , C , Δ are fixed parameters for all tokens in this sequence
- Every token in the sequence use the same way to update h



Click on the arrows to update the hidden state

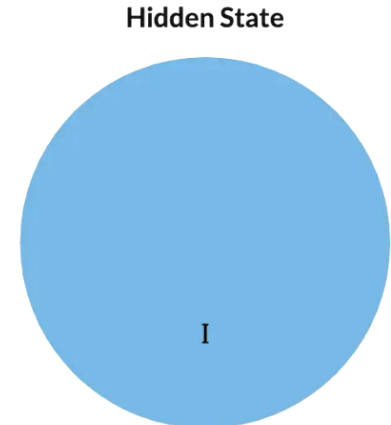
I want to order a hamburger



(This is an oversimplification, but it should give you a sense of what's going on.)

Click on the arrows to update the hidden state

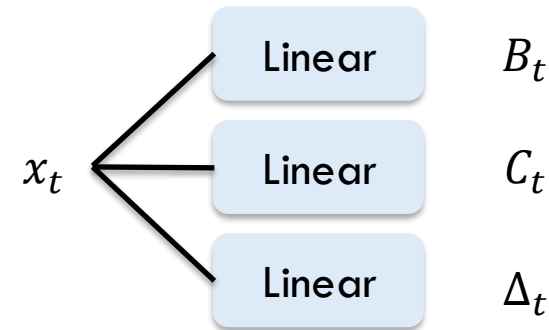
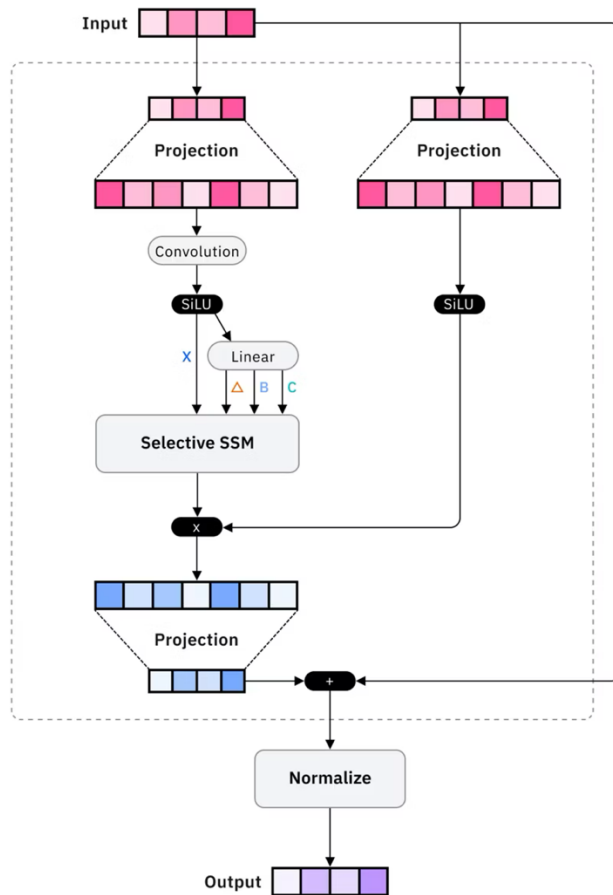
I want to order a hamburger



(Also an oversimplification.)

Mamba – Selective State Space Models (S6)

2. Mamba Block



Algorithm 2 SSM + Selection (S6)

Input: $x : (B, L, D)$

Output: $y : (B, L, D)$

1: $A : (D, N) \leftarrow \text{Parameter}$

▸ Represents structured $N \times N$ matrix

2: $B : (B, L, N) \leftarrow s_B(x)$

3: $C : (B, L, N) \leftarrow s_C(x)$

4: $\Delta : (B, L, D) \leftarrow \tau_\Delta(\text{Parameter} + s_\Delta(x))$

5: $\bar{A}, \bar{B} : (B, L, D, N) \leftarrow \text{discretize}(\Delta, A, B)$

6: $y \leftarrow \text{SSM}(\bar{A}, \bar{B}, C)(x)$

▸ **Time-varying:** recurrence (*scan*) only

7: **return** y

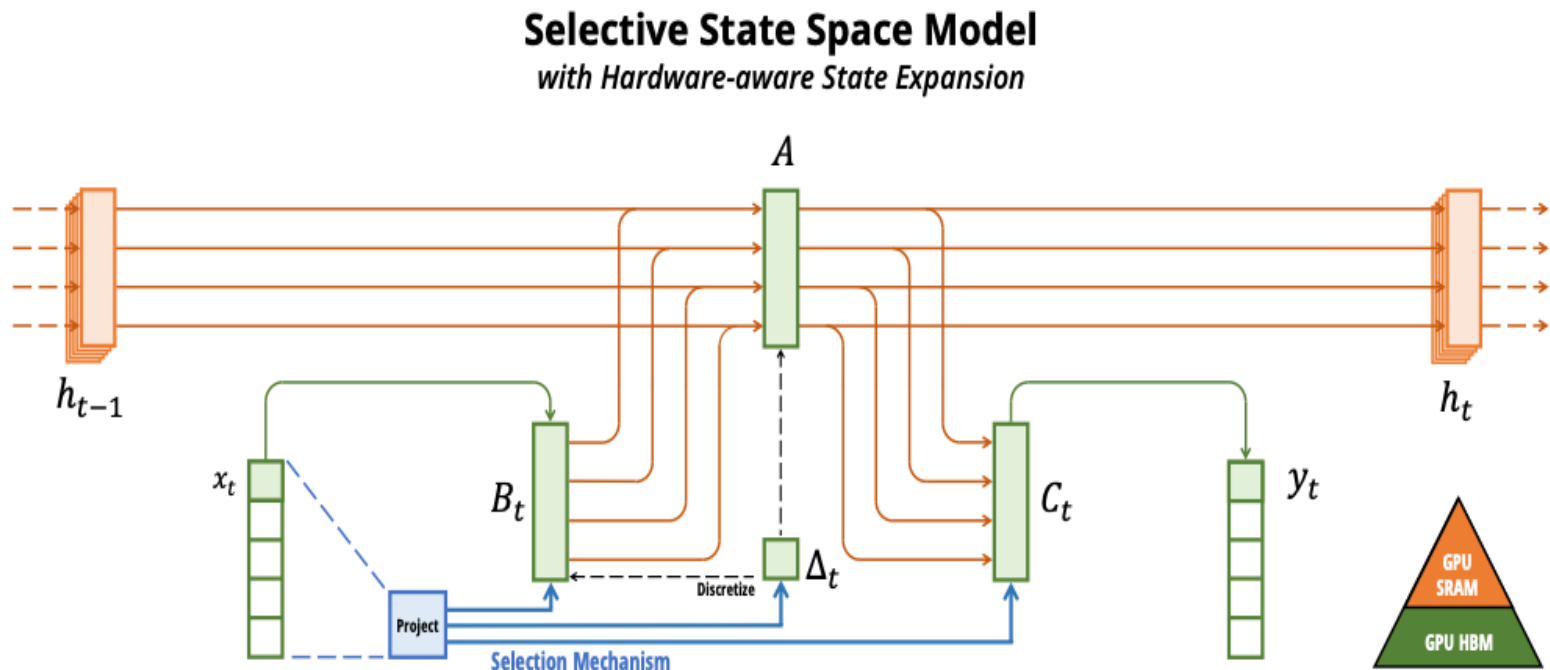
S4's Kernel

$$\mathbf{K} = (\bar{C}\bar{A}^{L-1}\bar{B}, \bar{C}\bar{A}^{L-2}\bar{B}, \dots, \bar{C}\bar{A}\bar{B}, \bar{C}\bar{B})$$

Mamba can not use convolutional trick anymore.

Mamba – Selective State Space Models (S6)

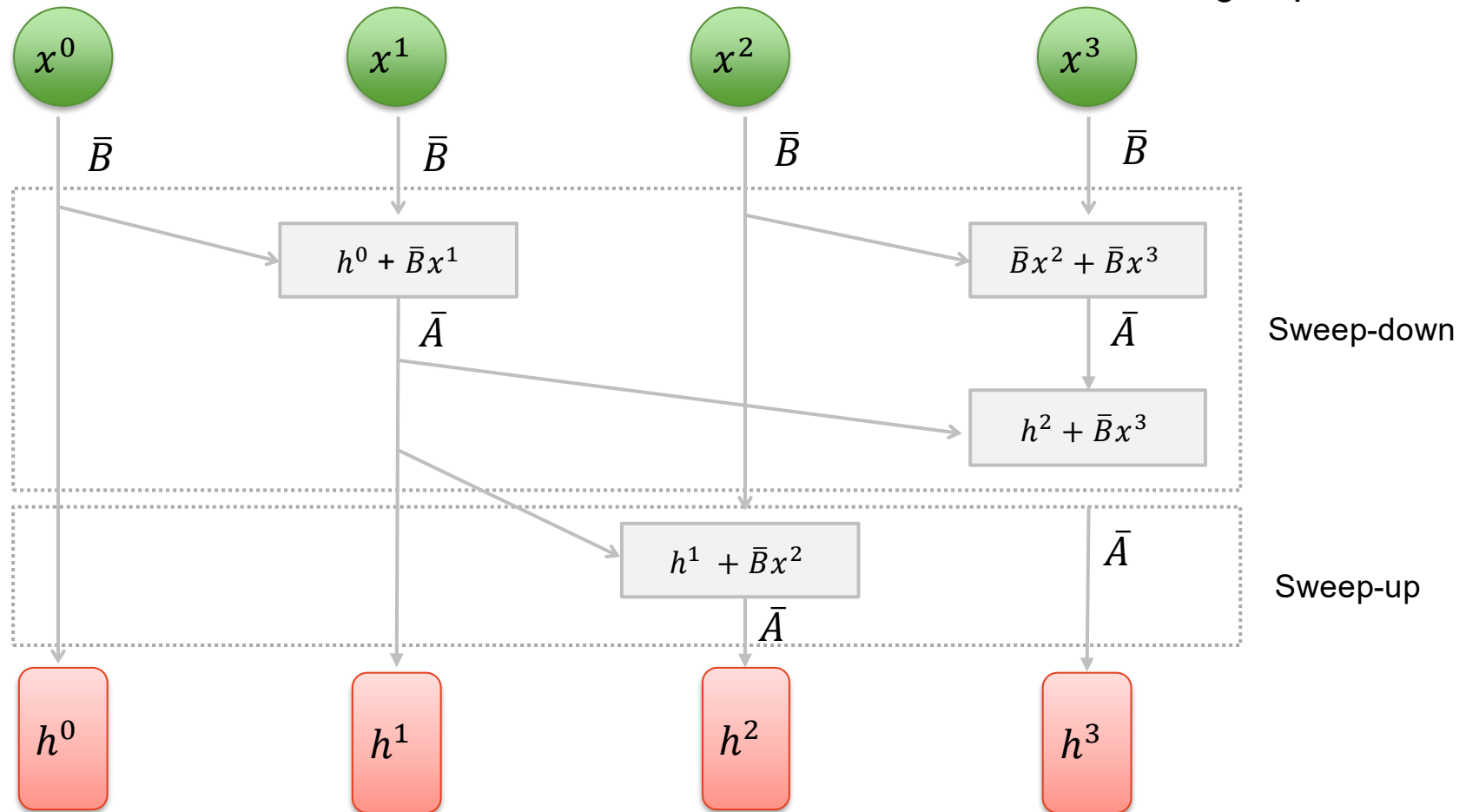
3. Hardware-aware State Expansion



IO is a big problem in Mamba model

Mamba – Selective State Space Models (S6)

4. Parallel Scan

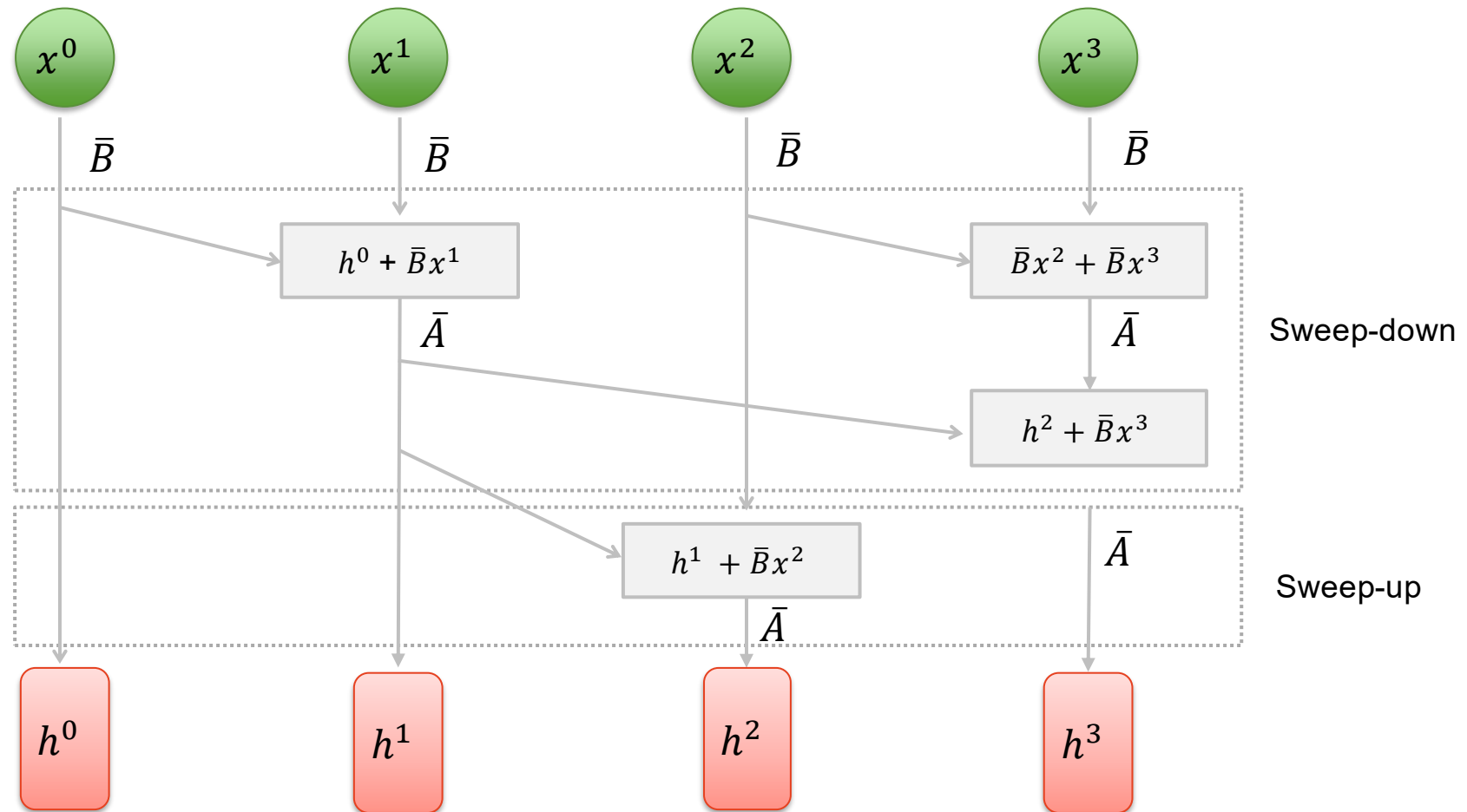


This process allows for some parallel computation of intermediate steps.

Mamba – Selective State Space Models (S6)

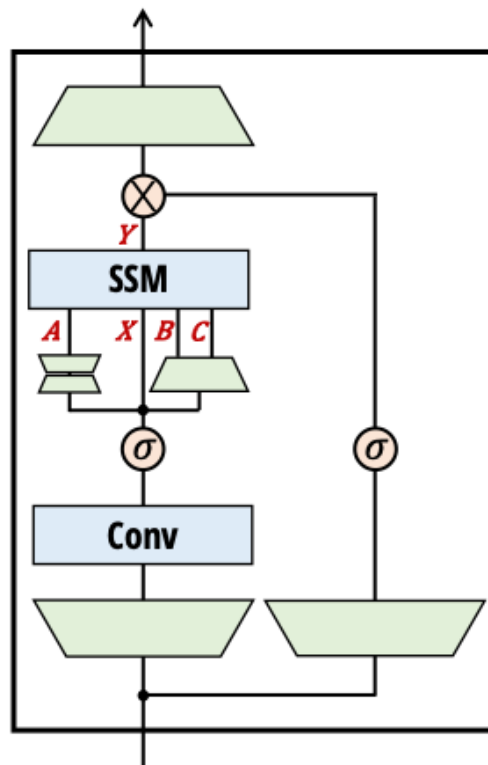
Can RNN do Parallel Scan as well?

$$\square h_t = \tanh(w_{hh}h_{t-1} + w_{hx}x_t + b_h)$$

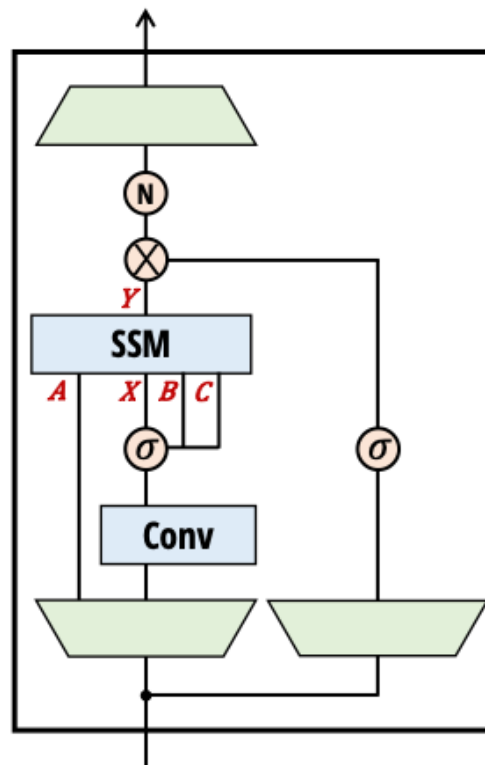


Parallel scan only works when the update is linear and associative.

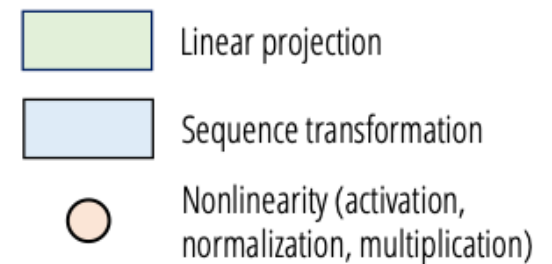
Mamba 2



Sequential Mamba Block

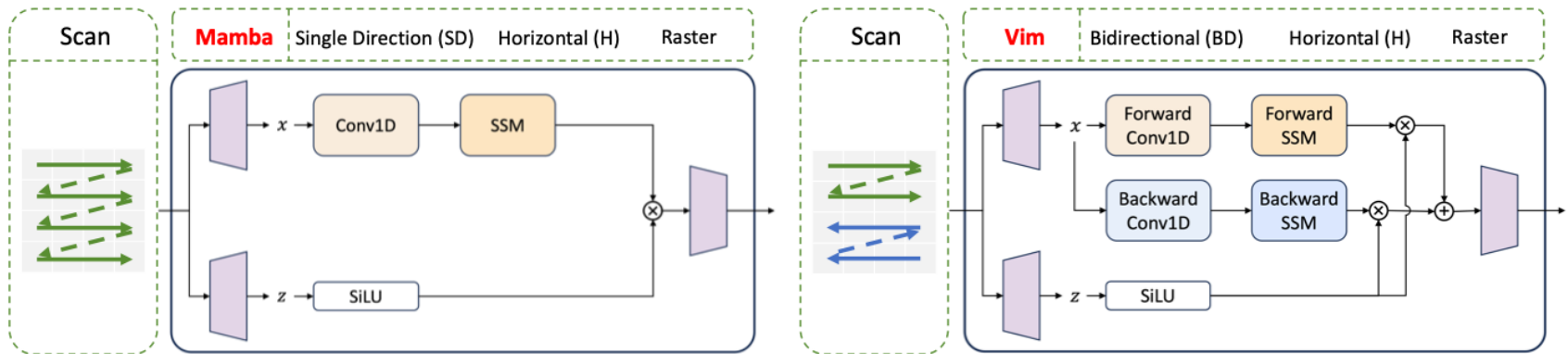
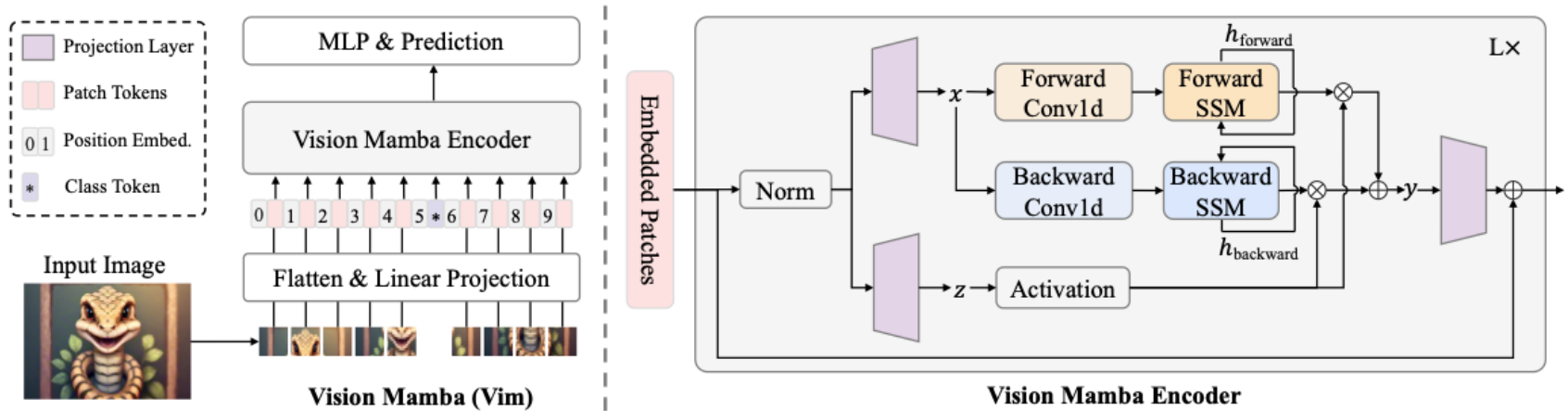


Parallel Mamba Block



Mamba for Vision

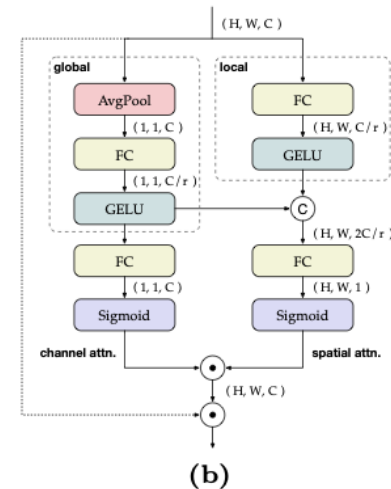
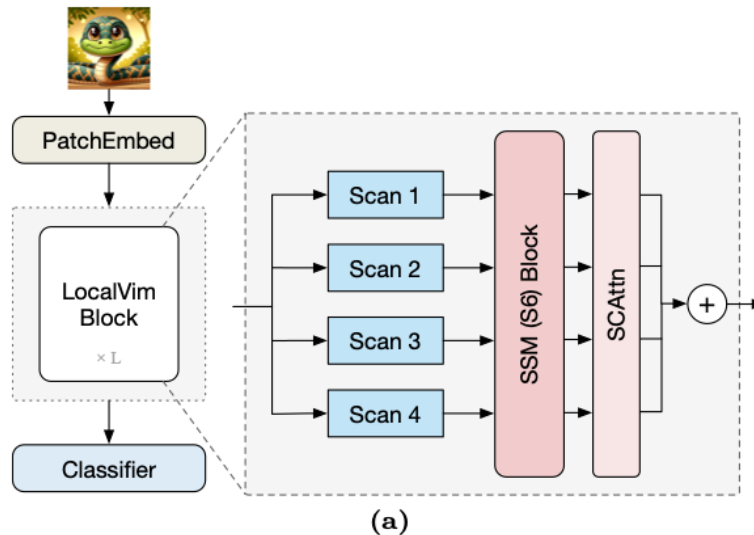
Vision Mamba



Zhu et al., Vision Mamba: Efficient Visual Representation Learning with Bidirectional State Space Model, **ICML 2024** (Citations: 3000+)

Mamba for Vision

LocalMamba



Mamba for Vision

Scanning Scheme in Vision Mamba Backbones

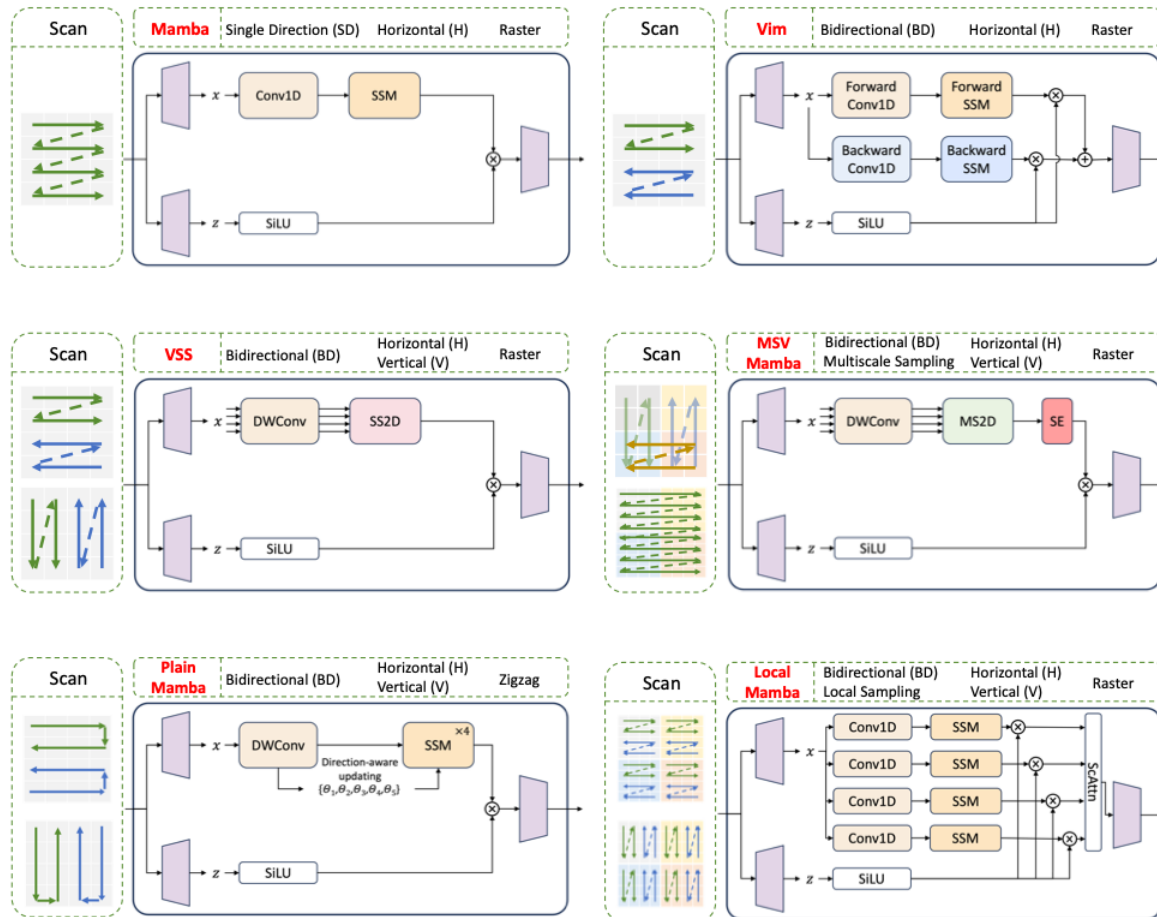


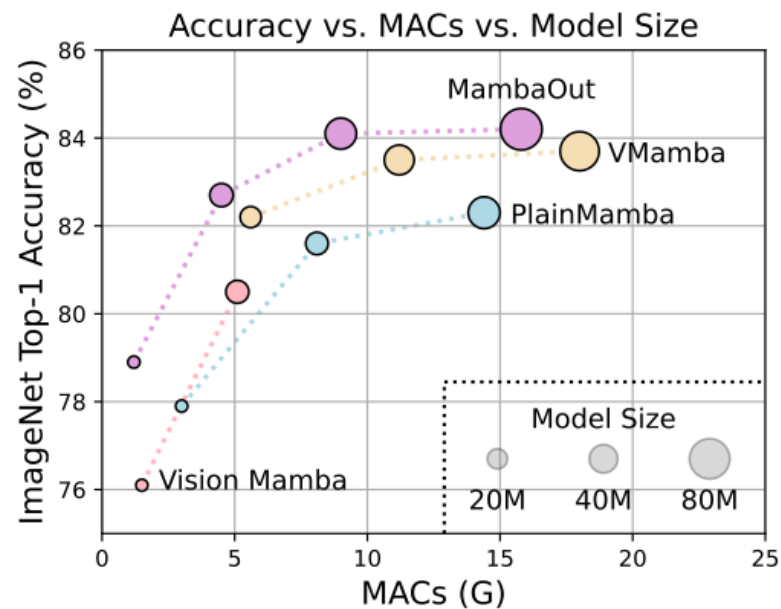
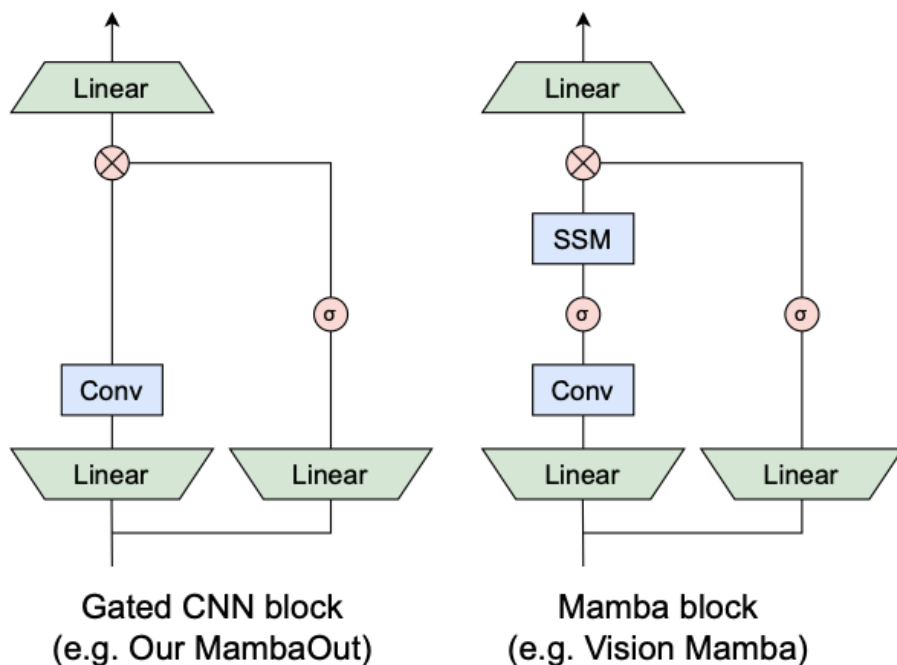
Figure borrowed from “Visual Mamba: A Survey and New Outlooks” <https://arxiv.org/html/2404.18861v2>

MambaOut: Do We Really Need Mamba for Vision?



In memory of Kobe Bryant

“What can I say, Mamba out.” — Kobe Bryant’s NBA farewell speech in 2016.



MambaOut: Do We Really Need Mamba for Vision?, CVPR 2025

Thank you !